

Evolutionary insights into the stereoselectivity of imine reductases based on ancestral sequence reconstruction

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I. Protein sequences, site-directed mutagenesis, expression, and purification

General information

All ancestral genes were synthesized and provided in plasmid form by Genscript Corporation (Nanjing, China). The synthetic genes were translated into competent *E. coli* BL21 (DE3). The DNA polymerase was purchased from Takara Biotechnology Company Limited (Dalian, China) and the restricted enzymes were purchased from New England BioLabs (NEB). All primers were purchased from Tsingke Biotechnology Company Limited (Shanghai, China).

Plasmid and protein sequences

N1

The ancestral gene encoding N1 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an *N*-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N1

```
atgggcagcagccatcatcatcatcatcacagcagcgccctggcgccgcggcagccatATGTCAAATAATAGTCCGGTG
ACCGTTATTGGTCTGGGTCAGATGGGCTCAGCCTTAGCAGGCGCATTCTGAAAGCAGGT
CATCCGACCACCGTTTGAATCGTACTCCTGGTAAAGCAGATCCGTTAGTGGCACGCGGC
GCCGTTTCGCGCCGCAACCGTTGCCGAAGCAGTGGCCGCCTCTCCGTTAGTTGTTGTGTGT
GTGAGCGATTATGATGCAGTTCGCGAAGTGTTAGATCCGGTTGCAGATGCCCTGGCAGGT
CGTGTGCTGGTGAATCTGACCTCAGGCACCCCGGAACAGGCACGCGAAATGGCCGCATG
GGCCGCCGGTCACGGCGCCGATTATCTGGATGGTGCAATTATGGCAACCCCGCGCTTAAT
TGGTACGCCGGAAGCCCTGATTCTGTATAGCGGTAGTCCGGAAGCCTTTGAAGCCCATCG
TGAAACCTTAGCCGCATTAGGTGATAGTACCTATTTAGGTGCCGATCCGGGGCTTAGCCTC
TCTGTATGATACCGCACTGCTGGGTTTAATGTGGGGGACGCTGAGCGGCTTTTTACATGC
AGCAGCACTGGTTGGTGCAGAAGGCGTGGATGCCGCAGCATTTGCCCCGGTTGCCACCGC
CTGGCTGAGCGCCGTGAGCTTTATGACCGATTATGCCGAACAGATTGATGCAGGTCGCTA
TCCGGGCGATGCCACCTTAGATGTTTCATGCCGCAGCAATTGATCATTTAATTCATGCAAG
TCGTGCACGTGGTATTGATACCGATTACCAGGAACTGTTTAAAGCCTTACTGGAACGCGC
CGTGGCAGCAGGGCATGGCGGCGATTCTTTTGCATCAGTTATTGAAGTGCTGCGTAAACC
GGCC
```

Amino acid sequence of N1

```
MGSSHHHHHHSSGLVPRGSHMSNNSPVTVIGLGQMGSALAGAFKAGHPPTTVWNRTPGKA
DPLVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLDPVADALAGRVLVNLTSGTPEQ
AREMAAWAAGHGADYLDGAIMATPRLIGTPEALILYSGSPEAFEHRETLAALGDSTYLAD
PGLASLYDTALLGLMWGTLGFLHAAALVGAEGVDAAAFAPVATAWLSAVSFMTDYAEQI
DAGRYPGDATLDVHAAIDHLIHASRARGIDTDLPELFKALLERAVAAGHGGDSFASVIEVL
RKPA
```

N2

The ancestral gene encoding N2 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N2

```
atgggcagcagccatcatcatcatcatcacagcagcgccctgggtgccgcggcgagccatATGAGTAATAATACACCAGTG
ACAGTTATCGGTTTtaggtcagatgggctcagccttagcggggcgcatttctgaaagcaggt
CATCTACGACCGTTTggaatcgtaccccaggtaaagccgattccactggttgctcgtggcg
CCGTTcgcgccgcaacagtgggcgaagctgtggcagctagtccattagttgttgtttgtg
TGTCAGATTATGATGCCGTTcgcgaagtTTTAGACCCGGTTGCGGATGCCTTAGCAGGTC
GCGTGTTAGTGAATCTGACGAGCGGCACACCAGAACAGGCACGCGAAATGGCAGCTTGG
GCGGCAGGGCACGGTGCCGATTATCTGGATGGCGCCATCATGGCTACACCTCCTATGATC
GGTACACCAGAAGCATTAACTCTGTATAGCGGTAGTCCAGAAGCCTTTGAAGCACATCGT
GAAACGTTAGCCGCACTGGGTGATTCTACATATCTGGGTGCAGATCCGGGTCTGGCCTCA
CTGTATGATACCGCTCTGTTAGGCTTAATGTGGGGTACGCTGACGGGCTTTTTGCACGCC
GCTGCCCTGGTTGGTGCCGAAGGCGTTCCAGCAGCTGCATTTGCTCCGGTTGCAACCGCT
TGGCTGAGTGCTGTGTCTTTTATGACGGAATATGCTGAACAGATCGATGCGGGTCGCTAT
CCGGGCGATGCAACCTTAGATGTTTATTAGCGGCAATTGATCATTTAATTCATGCATCTC
GTGCACGTGGTATTGATACGGCCTTACCGGAATTATTAAAGCCTTACTGGAACGCGCAG
TTGCAGCTGGACACGGTGGTGATAGTTTTCGAGCGTTATTGAAGTGTTACGTAAACCTG
CG
```

Amino acid sequence of N2

```
MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGQMGSALAGAFKAGHPPTTVWNRTPGKA
DPLVARGAVRAATVAEVAASPLVVVCVSDYDAVREVLDPVADALAGRVLVNLTSGTPEQ
AREMAAWAAGHGADYLDGAIMPMPMIGTPEALILYSGSPEAFEHRETLAALGDSTYLGA
DPGLASLYDTALLGLMWGTLTGFLHAAALVGAEGVPAAAFAPVATAWLSAVSFMTEYAEQI
DAGRYPGDATLDVHLAAIDHLIHASRARGIDTALPELFKALLERAVAAGHGGDSFASVIEVLR
KPA
```

N3

The ancestral gene encoding N3 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N3

```
atgggcagcagccatcatcatcatcatcacagcagcgccctgggtgccgcggcgagccatATGTCAAATAATACCCCTGTGA
CAGTGATCGGTCTGGGCTTAATGGGTAGCGCACTGGCCGGCGCCTTTCTGAAAGCAGGTC
ATCCAACGACCGTTTggaatcgtacgccaggtaaagccgacccattagttgctcgtgggtg
CAGTTCGTGCAGCAACCGTGGCAGAAGCCGTTGCTGCCTCTCCTCTGGTTGTGGTTTGTGT
GAGCGATTATGATGCGGTTcgcgaagtGTTAGATCCAGTTGCGGATGCCCTGGCGGGTTCG
CGTGTTAGTGAATCTGACGTCAGGCACCCCGGAAGAAGCTCGCGAAACGGCAGCTTGGG
CCGCTGAACATGGTGCGGATTATTTAGATGGCGCTATCATGGCCACACCTCCTATGATCG
GTACACCAGAAGCCTTAATCTTATATTCAGGTAGTCCGGAAGCATTTGAAGCCCATCGTG
```

AAACCTTAGCTGCATTAGGTGGTTCTACCTATTTAGGCGCCGATCCGGGTTTAGCAAGTC
TGTATGATGTTGCACTGTTAGGCTTAATGTGGGGCACCCCTGGCAGGCTTTTACATGCCGC
TGCCTTAGTGGGCTCAGAAGGCGTTCCAGCCGCTGCGTTTGCTCCGTTTGCTACAGCTTGG
CTGAGTGCTGTGTCTTTTCTGACCGAATATGCCGAACAGATTGATGCGGGCGATTATCCG
GGCGATGCTACACTGGATACACATCTGGCCGCAATTGAACATTTAATTCATGCAAGCCGC
GCACGCGGTATCGATACGGCCCTGCCAGAAGTGTAAAAAGCATTACTGGAACGCGCAGTT
GCGGCAGGTCATGGTGGTGATAGCTTTCATCAGTTATCGAAGTGTTACGTAAACCAGCG

Amino acid sequence of N3

MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGLMGSALAGAFKAGHPPTTVWNR
TPGKADPLVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLPVADALAG
RVLVNLTSGTPEEARETAAWAAEHGADYLDGAIMATPPMIGTPEALILYSGSPEA
FEAHRETLAALGGSTYLGADPGLASLYDVALLGLMWGTLAGFLHAAALVGSEG
VPAAAFAPFATAWLSAVSFLTEYAEQIDAGDYPGDATLDTHLAAIEHLIHASRAR
GIDTALPELLKALLERAVAAGHGGDSFASVIEVLRKPA

N4

The ancestral gene encoding N4 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N4

atgggcagcagccatcatcatcatcatcacagcagcgccgtggtgccgcggcgagccatATGAGTAATAATACGCCAGTG
ACGGTGATCGGCTTAGGTTTAATGGGCTCAGCCTTAGCGGGCGCCTTTCTGAAAGCAGGT
CATCCGACGACCGTTTGAATCGCACACCAGGTAAAGCCGATCCGTTAGTTGCTCGCGGC
GCTGTTCGTGCTGCCACAGTGGCCGAAGCTGTGGCAGCTAGTCCTCTGGTTGTGGTTTGT
GTGTCAGATTATGATGCAGTTCGCGAAGTTCTCGATCCAGTTGCGGATGCCCTGGCCGGT
CGCGTGTTAGTGAATCTGACCTCAGGTACACCAGAAGAAGCTCGCGAAACCGCTGCTTGG
GCTGCGGAACATGGTGCAGATTATTTAGATGGTGCAATCATGGCAACCCCTCCTATGATC
GGTACACCGGAAGCGTTAATCTTATATAGCGGTTACGCGAAGCATTGTAAGCCCATCGT
GAAACGTTAGCAGCCCTGGGTGGTAGTACCTATCTGGGTGAAGATCCAGGCTTAGCCGCA
CTGTATGATGTTGCCTTACTGGGTTTAATGTGGGGAACCCTGGCAGGCTTTTACATGCCG
CCGCATTAGTTGGTTCAGAAGGCGTTCCTGCGGCAGCATTGCTCCATTTGCAACGGCTT
GGCTGAGCGCCGTGTCTTTTCTGACAGAAATGGCCGAACAGATTGATGCAGGCGATTATC
CGGGCGATGCTACACTGGATACCCATGCGGCAGCTATCGAACATTTAATTCATGCGTCAC
GTGCACGTGGTATTGATACGGCCTTACCGGAAGCTCTGAAAGCACTGTTAGAACGCGCTG
TTGCTGCCGGTCATGGTGGCGATTCTTTTGCATCAGTTATTGAAGTGTTACGTAAACCAGC
G

Amino acid sequence of N4

MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGLMGSALAGAFKAGHPPTTVWNRTPGKA
DPLVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLPVADALAGRVLVNLTSGTPEE
ARETAAWAAEHGADYLDGAIMATPPMIGTPEALILYSGSREAFEAHRETLAALGGSTYLGED
PGLAALYDVALLGLMWGTLAGFLHAAALVGSEGVPAAAFAPFATAWLSAVSFLTEMAEQID
AGDYPGDATLDTHAAIEHLIHASRARGIDTALPEALKALLERAVAAGHGGDSFASVIEVLR
KPA

N5

The ancestral gene encoding N5 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N5

```
atgggcagcagccatcatcatcatcatcacagcagcggcctgggtgccgcgcggcagccatATGTCAAATAATACACCAGTG
ACCGTTATCGGCTTAGGCTTAATGGGTAGCGCTTTAGCGGGCGCCTTTCTGAAAGCCGGT
CATCCGACGACAGTTTGGAATCGTACGCCAGGTAAAGCCGATCCATTAGTGGCACGTGGT
GCAGTTCGCGCAGCTACCGTTGCCGAAGCTGTTGCTGCGAGTCCTCTGGTTGTTGTGTGTG
TGAGCGATTATGATGCTGTTTCGTGAAGTTTATAGATCCGGTTGCGGATGCCCTGGCGGGTC
GCGTGTTAGTTAATTTAACGTCAGGGACACCGGAAGAAGCTCGCGAAACCGCCGCTTGG
GCCGCTGAACATGGTGCAGATTATCTGGATGGTGCAATCATGGCCACACCTCCTATGATC
GGTACGCCTGAAGCCTTAATCTTATATTACAGTTTCTCGCGAAGTGTTTGAAGCCCATCGT
GAAACACTGGCCGCACTGGGTGGCTCTACATATTTAGGCGAAGATCCAGGTTTATAGCAGCC
CTGTATGATGTTGCCCTTACTGGGTCTGATGTGGGGCACCTTAGCAGGCTTTCTTCATGCGG
CAGCGTTAGTTGGCTCAGAAGGCGTTCTGCTGCGGAATTTGCTCCATTTGCCACGGCTT
GGTTATCAGCAGTGTCTTTTCTGCCAGAAATGGCAGAACAGATCGATGCCGGCGATTATC
CGGGCGATAGTACACTGGCTATGAATGCAGCCGCAATCGAACATCTGATTCATGCAAGTC
GTGCACAGGGTATTGATACGGCATTACCAGAAGCATTAAAAGCATTACTGGAACGCGCT
GTGGCGGCCGGTCATGGTGGTGATAGCTTTGCAAGTGTTATTGAAGTTTACGTAAACCT
GCT
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Amino acid sequence of N5

```
MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGLMGSALAGAFKAGHPPTVWNRTPGKA
DPLVARGAVRAATVAEVAASPLVVVCVSDYDAVREVLDPVADALAGRVLVNLTSGTPEE
ARETAAWAAEHGADYLDGAIMATPPMIGTPEALILYSGSREVFEAHRETLAALGGSTYLGED
PGLAALYDVALLGLMWGTLAGFLHAAALVGSEGVPAAEFAPFATAWLSAVSFLPEMAEQID
AGDYPGDSTLAMNAAIEHLIHASRAQGIDTALPEALKALLERAVAAGHGGSFASVIEVLR
KPA
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N557

The ancestral gene encoding N557 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N557

```
atgggcagcagccatcatcatcatcatcacagcagcggcctgggtgccgcgcggcagccatATGAGTAATAATACCCCGGTG
ACCGTTATTGGCTTAGGTCCGATGGGCTCTGCACTGGCAGGCGCCTTTCTGGCCGCAGGT
CATCCGACCACCGTTTGGAATCGAACCCTGGTAAAGCAGATCCGTTAGTTGCACGCGGC
GCAGTTCGCGCCGCAACCGTGCCGAAGCAGTGGCAGCCTCTCCGTTAGTTGTTGTGTGTG
GTGAGCGATTATGATGCAATGCGTGAAGTGCTGGACCCGGTTGCCGATGCCCTGGCAGGT
CGTGTGCTGGTTAATCTGACCTCAGGGACCCCGGAAGAAGCACGTGAAGCAGCCGCTG
GGCCGCCGAACATGGTGCCGATTATCTGGATGGTGCCATTATGGCAACCCCGCCGATGAT
TGGCACTCCGGAAGCCTTAATTTTTTATAGCGGTCCGCGCGAAGTGTTTGAAGCCCATCG
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TGAAACCCTGGCAGCATTAGGCGGCTCTACCTATTTAGGCGAAGATCCGGGCTTAGCAGC
ACTGTATGATGTTGCATTACTGGGTTTAATGTGGACCACCTTAGCCGGCTTTTACATGCC
GCAGCACTGGTGGGTTCAGAAGGCGTTCCGGCCGCAGAATTTGCACCGTTTGCCACCGCA
TGGTTTAGTGCCGTGTCTTTTCTGCCGGAATGGCAGAACAGATTGATGCAGGCGATTAT
CCGGGCGATAGTACCTTAGCCATGAATGCAGCAGCCATTGAACATCTGATTCATGCAAGT
CGCGCCAGGGTATTGATACCGCCTTACCGAAGCCCTGAAAGCATTAGCCGAACGCGCA
ATTCCGCGAGGTATGGCGGTGATAGCTTTGCCTCAGTTATTGAAGTGCTGCGTAAACCG
GCC

Amino acid sequence of N557

MGSSHHHHHHSSGLVPRGSHMSNNTPTVTIGLGPMGSALAGAFLAAGHPPTTVWNRTPGKAD
PLVARGAVRAATVAEAVAASPLVVVCVSDYDAMREVLDPVADALAGRVLVNLTSGTPPEA
REAAA WAAEHGADYLDGAIMATPPMIGTPEALIFYSGPREVF EAHRETLAALGGSTYLGEDP
GLAALYDVALLGLMWTTLAGFLHAAALVGSEGVPAAEFAPFATAWFSAVSFLPEMAEQIDA
GDYPGDSTLAMNAAAIEHLIHASRAQGIDTALPEALKALAERIAAAGHGGSFASVIEVLRKP
A

N558

The ancestral gene encoding N558 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N558

atgggcagcagccatcatcatcatcacagcagcggcctgggtgccgcggcgagccatATGTCAAATAATACCCCGGTG
ACCGTTATTGGTCTGGGCCCCGATGGGTAGTGCCTTAGCAGGCGCCTTTCTGGCCGCAGGT
CATCCGACCACCGTTTGGAATCGTACTCTGGTAAAGCCGATCCGTTAGTTGCACGCGGT
GCCGTTCTGTGCCGCCACCGTGGCCGAAGCAGTTGCAGCCTCTCCGTTAGTTGTTGTGTGT
GTGAGCGATTATGATGCAATGCGTGCAGTTCTGGACCCGGCCGCAGATGCCCTGGCAGGT
CGCGTGCTGGTTAATCTGACCTCAGGTACGCCGAAGAAGCACGTGAAGCAGCCGCATG
GGCAGCAGAACATGGCGCAGATTATCTGGATGGTGCCATTATGGTTCCGCCGCCGATGAT
TGGAACCCCGGAAGCCTTAATTTTTTATAGCGGTCCGCGCGAAGTGTTGAAGCCCATCG
CGCAACCTTAGCAGCATTAGGCGGCTCTACCTATTTAGGCGAAGATCCGGGTTTAGCAGC
ACTGTATGATGTGGCACTGTTAGGCTTAATGTGGACCACCATGAGCGGCTTTTTACATGC
AGCCGCATTAGTGGGCTCAGAAGGTGTTCCGGCAGCCGAATTTGCACCGTTTGCCACCGC
CTGGTTTTCTAGCGTGAGCTTTCTGGAAGAAATGGCCGAACAGATTGATGCAGGCGATTA
TCCGGGCGATGAATCAACCCTGGCAATGAATGCCGCAGCCATTGAACATATTATTCATGC
AAGTCGCGCCCAGGGTATTGATACCGCCTTACCGGAAGCCCTGAAAGCACTGGCCGAAC
GCGCAATTGCCGCAGGTCATGGTGAAGATAGCTTTGCATCAGTGATTGAAGTGTTACGTA
AACCGGCA

Amino acid sequence of N558

MGSSHHHHHHSSGLVPRGSHMSNNTPTVTIGLGPMGSALAGAFLAAGHPPTTVWNRTPGKAD
PLVARGAVRAATVAEAVAASPLVVVCVSDYDAMRAVLDPAADALAGRVLVNLTSGTPPEA
REAAA WAAEHGADYLDGAIMVPPPMIGTPEALIFYSGPREVF EAHRA TL AALGGSTYLGEDP
GLAALYDVALLGLMWTTMSGFLHAAALVGSEGVPAAEFAPFATAWFSSVSFLEMAEQIDA
GDYPGDESTLAMNAAAIEHHIHASRAQGIDTALPEALKALAERIAAAGHGGSFASVIEVLRKP
A

N559

The ancestral gene encoding N559 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N559

```
atgggcagcagccatcatcatcatcatcacagcagcggcctggcgccgcggcagccatATGTCTAATAATACCCCGGTGA  
CCGTGATTGGTTTAGGTCCGATGGGTTCGCGCACTGGCAGGCGCATTTCTGGCAGCCGGTC  
ATCCGACCACCGTGTGGAATCGTACACCGGGTAAAGCAGATGCCTTAGTGGCACGCGGT  
GCAGTTCGTGCAGCCACCGTTGCCGAAGCAGTGGCCGCTCTCCGTTAGTTGTTGTTTGTG  
TGACCGATTATGATGCCATGCGTGCAATTCTGGAACCGGCAGCCGATGCATTAGCAGGTC  
GTGTGTTAGTTAATCTGACCTCAGATACCCCGGAACGCGCACGCGAAGCCGCCGCATGGG  
CCGCAGAACATGGTGCCGATTATCTGGATGGTGCAATTATGGTTCCGCCGCCGATGATTG  
GTACTCCGGAAGCCTTAATTTTTTATAGCGGTCCGCGTGAAGTGTTTGAAGCACATCGCG  
CAACCCTGGCAGCATTAGGCGGCTCTACCTATCTGGGCGAAGATCCGGGCTTAGCCGCCC  
TGTATGATGTTGCACTGCTGGATCTGTTTTGGACCACCATGAGCGGCTATTTACATGCAGC  
AGCCTTAGTGGGCTCAGAAGGCGTTCGCGCAGCAGAATTTGCCCCGTTTGCCACCGCCAT  
TTTTGATTCACTGAGCTTTCTGGAAGAAATGGCCGAACAGATTGATGCAGGCGAATATCC  
GGGCGATGAATCAACCTTAGCAATGAATGCCGCAGGTATTGATCATATTATTCATGCAAG  
TCGCGCACAGGGTATTGATACCGCCTTACCGGAAGCCGTTAAAGCCCTGGCCGAACGTGC  
CATTGCAGCAGGTCATGGTGAAGATAGCTTTACCTCAGTTATTGAAGTGCTGCGTAAACC  
GGCC
```

Amino acid sequence of N559

```
MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGPMGRALAGAFLAAGHPTTVWNRTPGKA  
DALVARGAVRAATVAEAVAASPLVVVCVTDYDAMRAILEPAADALAGRVLVNLTSDTPER  
AREAAAWAAEHGADYLDGAIMVPPPMIGTPEALIFYSGPREVFEAHRATLAALGGSTYLGED  
PGLAALYDVALDLFWTTMSGYLHAAALVGSEGVPAAEFAPFATAIFDSLSFLEEMAEQIDA  
GEYPGDESTLAMNAAGIDHIIHASRAQGIDTALPEAVKALAERIAAAGHGEDSFTSVIEVLRKP  
A
```

N560

The ancestral gene encoding N560 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of N560

```
atgggcagcagccatcatcatcatcatcacagcagcggcctggcgccgcggcagccatATGAGTAATAATCAGCCGGTT  
ACCGTGATTGGCTTAGGTCCGATGGGTTCAGGCAATGGCAGGCGCCTTTCTGGCCGCAGGT  
CATCCGGTGACCGTGTGGAATCGTACCCCGTCTAAAGCAGATGCCTTAGTTGCCAGGGC  
GCAGTTCGTGCAGCCACCGTTGCCGAAGCATTAGCAGCCTCAGAATTAGTTGTGCTGTCA  
CTGACCGATTATGATGCAATGTATGCAATTCTGGAACCGGCCGCCGATGCCCTGGCAGGT  
CGCGTGTTAGTTAATCTGAGTAGCGATACCCCGGAACGCGCACGTGAAGCAGCCGCATG  
GGCAGCAGAACATGGTGCACGCTATCTGACAGGAGGCGTTATGGTTCCGCCGCCGGGTA  
TTGGAACACCGGAAGCGTATGTGTTTTATAGCGGTCCGCGCGAAGTGTTTGAAGCACATC
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GTGCCACCTTAGCAGTGCTGGGCGGCACCGATTATGTGGGTGAAGATCCGGGTCTGGCCC
AGCTGTATTATCAGGCACAGTTAGATATTTTTTGGACCACCATGACCGCCTATTTACATGC
AACCGCCCTGGTTGGCTCAGAAGGCGTTCCGGCCGAAGAATTTCTGCCGTATGCAACCGA
AATGTTTGATTCAATGAGCTTTCTGGAAGAAATGGCCGAACAGATTGATGCAGGCGAATA
TCCGGGCGATGAAGATACCTTAGCGATGGGCGCAGCCGGTGTGGATCATATTGTTTCATGC
CTCTCGCGATGCAGGTATTGATACCGCATTACCGGAAGCCGTTAAAGCCCTGTTTCGTCG
CGCCATTGCCGCAGGTCATGGTGAAGATAGCTTTACCTCACTGATTGAAGTGCTGCGTAA
ACCGGCC

Amino acid sequence of N560

MGSSHHHHHHSSGLVPRGSHMSNNQPVTVIGLGPMGQAMAGAFLAAGHPVTVWNRTPSKA
DALVAQGAVRAATVAEALAASELVVLSTDYDAMYAILEPAADALAGRVLVNLSSDTPERA
REAAAWAAEHGARYLTGGVMVPPPGIGTPEAYVFYSGPREVFEAHRATLAVLGGTDYVGED
PGLAQLYYQAQLDIFWTTMTAYLHATALVGSEGVPAEEFLPYATEMFDSMSFLEEMAEQIDA
GEYPGDEDTLAMGAAGVDHIVHASRDAGIDTALPEAVKALFRRRAIAAGHGEDSFTSLIEVLR
KPA

AltN1

The ancestral gene encoding AltN1 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN1

atgggcagcagccatcatcatcatcacagcagcggcctggcgccgcccgcagccatATGTCAAATAACACACCCGTTA
CTGTAATAGGCCTGGGCCAAATGGGTAGCGCACTCGCGGGCGCATTCTGGCGGCCGGTC
ATCCGACCACGGTTTGGAAATCGTACCCCGGCTAAAGCCGATGCACTGGTTGCGAGAGGC
GCGGTACGTGCTGCGACCGTGGCCGAGGCGGTTGCCGCCTCTCCGCTGGTTCGTGGTGTGC
GTTTCGGATTATGATGCGGTGCGCGAAGTTCTGGATCCGGTTGCGGACAGCCTGGCGGGC
CGCACCTTGGTCAACCTGACTTCAGGTACGCCAGAGGAGGCTCGTGAAACCGCAGCGTG
GGCAGCGTCCCATGGTGCGGACTACCTGGACGGCGCCATCATGAGCACCCTCCGATGGT
GGGCCGTCCGGAAGCTTTGTTCTGTACAGCGGTAGCCCTGAGGCTTTCGCAGCGCATCG
TGAGACCTTAGCTGCGTTGGGCGACTCCGTTACCTGGGTGCGGACCCGGGTTTAGCCGC
CTTGACGACACGGCTCTGTTCCGCCTGATGTGGGGTACACTTGCAGGCTTTCTGCACGC
AGTGGCGCTTGTGGGCGCGGAAGGTGTTCCGGCTGCTACCTTTGCGCCGGTGGCGACCGC
TTGGCTGCCGGCGGTGTCCCTGCTGACCGAATATGCACGCCAGATCGAGGCGGGTCGCTA
TCCGGGTGATGGTACGCTGGACGTTCACTTGGCGGCGATCGATCATTTGGTCCACACCAG
CCGTGCGCGTGGTATTGACACCGCCGTGCCAGAGCTGCTTAAGACCTTGTTGGACCGTGC
AGTTGCTGCGGGTCACGGCGCGGATTCTATTGCAAGCGTTATTGAAGTGCTGCGTAAAGC
AGCATAACTCGAG

Amino acid sequence of AltN1

MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLQMGSAFLAGAFLAAGHPPTTVWNRTPAKA
DALVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLDPVADSLAGRVLVNLTSGTPEEA
RETAAWAASHGADYLDGAIMSTTPMVGRPEALFLYSGSPEAFAAHRETLAALGDSVHLGAD
PGLAALYDTALFGLMWGTLAFLHAVALVGAEGVPAATFAPVATAWLPAVSLLTEYARQIE
AGRYPGDGTLDVHLAAIDHLVHTSRARGIDTAVPELLKTLLDRAVAAGHGADSIASVIEVLR
KAA

AltN2

The ancestral gene encoding AltN2 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN2

```
atgggcagcagccatcatcatcatcatcacagcagcggcctggcgccgcggcagccatATGAGTAATAACTCACCCGTA
ACAGTTATAGGCCTGGGTTCAGATGGGTCTGCGTTGGCGGGTGCCTTCCTGGCCGCAGGC
CACCCGACCACGGTTTGGAATCGTACCCCGGCAAAAGCAGACGCGCTGGTGGCGCGCGG
TGCCGTTTCGTGCGGCGACTGTAGCCGAAGCGGTGCGCGCTTCGCCGCTGGTGGTGGTTTG
CGTGTCCGATTACGATGCGGTGCGCGAGGTGCTGGACCCGGTTGCTGACAGCCTGGCAGG
GCGCACCTTGGTGAACCTGACTAGCGGTACACCGGAGGAGGCCCGTGAGACCGCGGCTT
GGGCAGCGTCCCATGGCGCGGACTACTTGGATGGTGCATCATGACCACCCCGCCTATGG
TTGGTCCCCGGAGGCTATGATCCTGTACAGCGGCAGCCGTGAGGCGTTTGAAGCTCATC
GTGAGACGTTGGCTGCTTTAGGCGACTCTGTTTATTTGGGCGCGGATCCGGGTCTCGCCG
CGCTTTACGACACCGCCCTGCTGGGTCTTATGTGGGGTACGCTGGCTGGTTTTCTGCATGC
TACCGCGTTGGTGGGAGCCGAAGGTGTGAGTGCAACCACCTTCGCCCCAGTTGCGACCGC
GTGGCTGAGCGCGGTGTCGTTCTGACCGATTATGCCCGCCAAATTGACGCGGGTGACTA
CCCGGGTGATGCGACGTTAGATGTACACGCAGCTGCGATCGATCACCTGGTTCACGCTAG
CCGTGCTAGAGGCATTGATACCGACCTGCCGGAAGTCTGAAGGCGCTCTTGGAACGTGC
AGTCGCAGCGGGCCACGGCGGTGACAGCTATGCAAGCGTTATTGAAGTTCTGCGTAAAG
CGGCGTAACTCGAG
```

Amino acid sequence of AltN2

```
MGSSHHHHHHSSGLVPRGSHMSNNSPVTVIGLGQMGSALAGAFLAAGHPTTVWNRTPAKA
DALVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLPVADSLAGRTLVLNLTSGTPEEA
RETAAWAASHGADYLDGAIMTTPPMVGSPEAMILYSGSREAFEAHRETLAALGDSVYLGAD
PGLAALYDTALLGLMWGTLAGFLHATALVGAEGVSATTFAPVATAWLSAVSFLTIDYARQID
AGDYPGDATLDVHAAIDHLVHASRARGIDTDLPELLKALLERAVAAGHGGSYASVIEVLR
KAA
```

AltN3

The ancestral gene encoding AltN3 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN3

```
atgggcagcagccatcatcatcatcatcacagcagcggcctggcgccgcggcagccatATGTCAAATAACACTCCCGTAA
CAGTTATAGGCCTGGGCCTTATGGGTAGCGCATTAGCAGGCGCGTTCTTGGCGGCTGGCC
ATCCGACCACCGTTTGGAATCGTACCCCGGGCAAGGCGGACGCTTGGTTCGCTCGCGGTG
CTGTTTCGTGCGGCGACGGTTGCAGAGGCTGTGGCGGCGTCTCCGCTGGTCGTGGTGTGCG
TTTCCGATTACGACGCGGTGCGTGAGGTGCTGGACCCGGTTGCCGACGCCTTGGCGGGTC
GTGTTCTGGTCAACCTGACCAGCGGCACCCCGGAGCAGGCACGCGAGATGGCGGCTTGG
GCAGCGGAGCACGGCGCGGACTACCTGGATGGTGCATTATGGCAACGCCACCGATGAT
TGGTACGCCGAGGCGGTGATTCTGTATAGCGTTCTCGTGAGGCCTTCGAAGCTCACCG
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TGAAACCCTCGCAGCGCTGGGTGGGTCCACCTATCTGGGTGCTGATCCGGGTCTGGCGGC
TCTGTACGATGTTGCGTTGCTCGGCATCATGTGGGGTACACTGAGCGGTTTTCTGCACGCC
GCTGCGTTGGTAGGTGCTGAAGGCGTTAGCGCGACCACTTTCGCCCTTTTGCTACTGCCT
GGCTGAGTGCGGTGTCGTTTCATGACCGATTACGCAAGACAAATTGACGCCGGTGAATATC
CGGGTGATGCGACGCTGGACACCCATGCAGCGGCCATCGAACACCTGGTTCATGCGAGC
CGCGCGCGCGGAATCGATACCGAACTGCCGGAATTTTTGAAGGCGTTGCTGGATCGTGCG
GTGGCAGCGGGCCACGGCGGTGACAGCTACGCATCCGTGATCGAGGTCCTGCGTAAACC
GGCATAACTCGAG

Amino acid sequence of AltN3

MGSSHHHHHSSGLVPRGSHMSNNTPTVTIGLGLMGSALAGAF LAAGHPPTTVWNRTPGKA
DALVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLDPVADALAGRVLVNLTSGTPEQ
AREMAAWAAEHGADYLDGAIMATPPMIGTPEAVILYSGSREAFEAHRETLAALGGSTYLGA
DPGLAALYDVALLGIMWGTL SGFLHAAALVGAEGVSATTFAPFATAWLSAVSFMTDYARQI
DAGEYPGDATLDT HAAAIEHLVHASRARGIDTELPEFLKALLDRAVAAGHGSDSYASVIEVL
RKPA

AltN4

The ancestral gene encoding AltN4 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN4

atgggcagcagccatcatcatcatcacagcagcggcctggcgccgagccatATGTCAAATAACACACCCGTTA
CTGTAATAGGCCTGGGCCTGATGGGTAGCGCATTGGCAGGCGCGTTTTGGCGGCTGGTC
ATCCGACCACCGTTTTGGAATCGTACCCCGGGTAAAGCAGACGCACTTGTGGCGCGTGGTG
CCGTTAGAGCGGCGACGGTTGCCGAGGCGGTTGCGGCGTCACCGCTGGTCGTGGTGTCG
TGAGCGATTACGACGCCGTGCGCGAGGTTCTAGACCCGTTGCGGATGCCCTCGCTGGTC
GTGTCTGGTCAACCTGACCAGCGGCACCCCTGAGCAGGCACGCGAGGCTGCTGCGTGG
GCAGCCGAGCACGGTGCGGACTATTTGGACGGCGCCATCATGGCGACGCCACCGATGAT
TGGTACTCCGGAAGCGCTGATTCTGTATAGCGGCTCTCGTGAGGTGTTCAAGCGCATCG
TGAGACACTGGCTGCGCTCGGCGTTTCGACCTACCTGGGCGAAGATCCAGGTCTGGCGTC
CCTGTACGACGTGGCTTTGCTGGGTATCATGTGGGGTACGCTGTCCGTTTTTTACATGCT
GCGGCGTTGGTGGGCGCCGAAGGTGTTCCGGCGGCCGAGTTCGCTCCGTTTTCGACCGCA
TGGCTGAGTGCAGTAAGCTTCCTGCCGATATGGCTCGCCAAATTGATTCCGGCGAGTAT
CCGGGTGATGCGACCTTGACATGCACGCAGCGGCGATCGAACACCTGGTGCACGCCTCT
CGTGCGCGCGGTATCGATACCGAACTGCCGGAAGCTTTGAAGGCGCTGCTGGATCGTGCG
GTGGCTGCGGGCCACGGCGCGGACAGCTTCGCAAGCGTTATTGAAGTTCTGCGTAAACCG
GCATAACTCGAG

Amino acid sequence of AltN4

MGSSHHHHHSSGLVPRGSHMSNNTPTVTIGLGLMGSALAGAF LAAGHPPTTVWNRTPGKA
DALVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLDPVADALAGRVLVNLTSGTPEQ
AREAAAWAAEHGADYLDGAIMATPPMIGTPEALILYSGSREVFEAHRETLAALGGSTYLGED
PGLASLYDVALLGIMWGTL SGFLHAAALVGAEGVPAAEFAPFATAWLSAVSFLPD MARQIDS
GEYPGDATLDMHAAAIEHLVHASRARGIDTELPEALKALLDRAVAAGHGADSFASVIEVLRK
PA

AltN5

The ancestral gene encoding AltN5 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN5

```
atgggcagcagccatcatcatcatcatcacagcagcgccctggcgccgcggcagccatATGTCAAATAACACACCCGTTA
CTGTAATAGGCCTGGGCCTGATGGGTAGCGCATTTGGCGGGCGCCTTCCTGGCTGCGGGTC
ACCCGACCACGGTGTGGAATCGTACCCCGGGCAAAGCGGACGCATTGGTTGCCCCGCGGT
GCCGTGCGTGCGGCTACCGTTGCGGAAGCTGTGCGGGCGTCGCCGCTGGTCGTGGTGTGC
GTTTCTGATTACGACGCGGTGCGCGAGGTGCTGGACCCAGTGGCGGACGCGTTGGCCGGT
AGAGTTCTGGTGAACCTCACCAGCGGCACCCCGGAGCAGGCTCGTGAGGCCGCTGCTTG
GGCAGCTGAGCACGGCGCGGACTACCTGGATGGTGTCTATCATGGCGACCCACCGATGA
TTGGTACGCCGAAGCGCTGATTCTGTACAGCGGTTCTCGTGAGGCATTCGAGGCGCATC
GTGAAACCTTAGCTGCATTAGGCGGTTTCGACTTATCTGGGCGAAGATCCGGGTCTGGCCG
CGTTGTATGATGTTGCGTTACTGGGTATCATGTGGTCCACGCTGTCCGGCTTTCTACACGC
GGCAGCGTTGGTGGGTAGCGAAGGTGTTCCGGCTGCTGAGTTCGCCCCTTTTGCGACCGC
ATGGCTGTCTGCGGTAAGCTTTCTGCCGGAGATGGCGCGTCAAATTGATAGCGGCGAATA
TCCGGGCGATAGCACTCTGGACATGAACGCAGCCGCCATCGAGCACCTCGTTCATGCATC
CCGCGCGCAGGGTATCGACACCGAACTGCCGGAAGCATTGAAGGCGCTGCTTGATCGCG
CGGTAGCTGCGGGCCATGGTGCGGACAGCTTCGCGAGCGTCATTGAAGTTTTGCGTAAAC
CGGCATAACTCGAG
```

Amino acid sequence of AltN5

```
MGSSHHHHHHSSGLVPRGSHMSNNTPVTVIGLGLMGSALAGAFLAAGHPTTVWNRTPGKA
DALVARGAVRAATVAEAVAASPLVVVCVSDYDAVREVLPVADALAGRVLVNLTSGTPEQ
AREAAAWAAEHGADYLDGAIMATPPMIGTPEALILYSGSREAFEAHRETLAALGGSTYLGED
PGLAALYDVALLGIMWSTLSGFLHAAALVGSEGVPAAEFAPFATAWLSAVSFLPEMARQIDS
GEYPGDSTLDMNAAAIEHLVHASRAQIDTELPEALKALLDRAVAAGHGADSFASVIEVLRK
PA
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AltN557

The ancestral gene encoding AltN557 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN557

```
atgggcagcagccatcatcatcatcatcacagcagcgccctggcgccgcggcagccatATGTCAAATAACACACCCGTTA
CTGTAATAGGACTGGGTCCGATGGGCAGCGCTTTGGCGGGTGCGTTTTTGGCCGCTGGTC
ATCCGACCACCGTTTTGGAATCGTACCCCGGGTAAAGCGGACGCATTGGTGGCTCGCGGTG
CGGTGCGTGCCGCGACTGTGGCGGAGGCCGTTGCGGCGTCGCCGCTGGTCGTCTGTGCG
TTACCGATTACGACGCCATGAGAGAGATTCTGGACCCGGCGGCGGATGCGTTGGCCGGTC
GTGTCCTGGTGAACCTGACCAGCGGCACGCCGAGCAGGCACGCGAGACTGCGGCCTGG
GCAGCGGAACACGGCGTGATTATCTGGACGGCGCGATCATGGCAACGCCACCGATGAT
TGGCACCCCGGAAGCTCTGATTTTCTATAGCGGTCCGCGTGAGGTTTTTGAAGCACATCG
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CGAAACCCTGGCGGCATTGGGCGGCAGCACGTACCTGGGCGAAGATCCGGGTCTGGCTG
CGCTGTATGACGTTGCGCTCTTAGGCATCATGTGGTCAACCATGTCTGGTTTCCTGCATGC
CGCTGCCCTCGTAGGTTCTGAAGGTGTTCCGGCAGAGGAATTCGCTCCGTTTGCTACCGC
GTGGCTTAGCGCAGTGAGCATTCTGCCTGAGATGGCGGCCAGATTGACTCCGGCGAATA
CCCGGGTGATTCCACCCTGGCTATGAACGCAGCAGCAATCGAGCACATCGTGACGCGA
GCCGTGCGCAAGGTATCGATACCGAACTGCCAGAGGCTTTGAAAGCGTTGGCGGAGCGC
GCGGTTGCAGCTGGCCACGGTGCGGACGGCTTCGCCTCCGTTATCGAGGTGCTGCGTAAG
CCGGCGTAACTCGAG

Amino acid sequence of AltN557

MGSSHHHHHSSGLVPRGSHMSNNTPTVTIGLGPMGSALAGAFLAAGHPTTVWNRTPGKAD
ALVARGAVRAATVAEAVAASPLVVVCVTDYDAMREILDPAADALAGRVLVNLTSGTPEQA
RETAAWAAEHGVLDGAIMATPPMIGTPEALIFYSGPREVFEAHRETLAALGGSTYLGEDP
GLAALYDVALLGIMWSTMSGFLHAAALVGSEGVPAEEFAPFATAWLSAVSILPEMAAQIDSG
EYPGDSTLAMNAAAIEHIVHASRAQGIDTELPEALKALAERA VAAGHGADGFASVIEVLRKP
A

AltN558

The ancestral gene encoding AltN558 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN558

atgggcagcagccatcatcatcatcacagcagcggcctggcgccgagcagccatATGACAAATAACACGCCCCTT
ACTGTAATAGGCCTGGGCCCTATGGGTAGCGCGTTGGCGGGTGCGTTTCTGGCTGCGGGT
CACCCGACCACCGTTTGAACCGTACCCCGGGTAAGGCGGACGCATTGGTAGCGCGCGG
TGCGGTTAGAGCCGCGACGGTGGCGGAGGCGGTGGCGGCCTCCCCGCTGGTGGTGGTGT
GCGTGACCGATTATGACGCAATGCGTGAAATCCTGGACCCGGCGGCTGACGCTCTCGCGG
GCCGTGTTCTGGTTAATCTGACCTCCGGCACCCAGAGCAGGCACGCGAGGCCGCCGCGT
GGGCAGCGGAACACGGTGTGATTACCTGGACGGGGCGATTATGGTTCCACCGCCGATG
ATTGGTACTCCGGAGGCTGTCGTGTTCTATAGCGGTCCGCGTGAGGTGTTTGAAGCGCAT
CGTGCCACGCTGGCCGCCTTGGGTGGCTCGACTTATCTGGGTGAAGATCCGGGCTTGGCG
GCGCTGTACGACGTGGCCTTGCTGGGCATCATGTGGTCAACGATGACCGGCTTCTGCAT
GCGGCTGCCTTAGTCGGTTCTGAAGGTGTTCCGGCTGAGGAATTTGCACCGTTCCGCCACC
GCATGGTTCAGCGCGGTAAGCATCCTTCCGGAGATGGCAGCTCAAATCGACAGCGGCGA
ATACCCGGGTGATGAAAGCACCCCTGGCAATGAACGCAGCGGCTATCGAGCACATCGTTC
ATGCTTCTCGTGATCAGGGTATTGATACCGAACTGCCGGAGGCGCTGAAAGCGCTGGCGG
AGCGCGCGGTGGCGGCTGGCCACGGCGCGGACGGCTTACCAGCGTCATTGAGGTTTTAC
GTAACCGGCATAACTCGAG

Amino acid sequence of AltN558

MGSSHHHHHSSGLVPRGSHMTNNTPTVTIGLGPMGSALAGAFLAAGHPTTVWNRTPGKA
DALVARGAVRAATVAEAVAASPLVVVCVTDYDAMREILDPAADALAGRVLVNLTSGTPEQ
AREAAWAAEHGVLDGAIMVPPPMIGTPEAVVFYSGPREVFEAHRATLAALGGSTYLGE
DPGLAALYDVALLGIMWSTMTGFLHAAALVGSEGVPAEEFAPFATAWFSAVSILPEMAAQID
SGEYPGDESTLAMNAAAIEHIVHASRDQIDTELPEALKALAERA VAAGHGADGFTSVIEVLR
KPA

AltN559

The ancestral gene encoding AltN559 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN559

```
atgggcagcagccatcatcatcatcatcacagcagcgccctgggtgccgcggcgagccatATGACAAATAACACGCCCCGTT
ACTGTAATAGGTTTATAGGCCCTATGGGTCAGGCACTCGCAGGCGCGTTCCTGGCGGCGGGC
CATCCGACCACCGTTTGAACCGTACCCCGGGGAAGGCAGACGAGCTGGTGGCACGCGG
TGCGGTGCGCGCGGCGACGGTGGCCGAGGCGGTGGCGGCCTCCCCGCTGGTTGTGGTGT
GCGTTACCGATTACGATGCAATGCGTGCTGTGTTGGAACCGGCTGCTGACGCTCTGGCGG
GCCGTGTTTTGGTCAACTTGACCTCCGATACCCCGGAGCGCGCGAGAGAAGCCGCGGCCT
GGGCAGCTGAGCACGGCGTAGATTACCTGGACGGCGCGATCATGGTTCCGCCACCGGTT
ATTGGCACCCCGGAAGCCGTGGTTTTCTATAGCGGTCCGCGTGAAGTTTTTCGAGGCACAT
CGTGCTACGCTGGCGGCGCTGGGTGGTGCACATATTTGGGCGAAGATCCGGGTCTGGCA
GCGCTGTATGACGTTGCGTTGCTGGACATCTTTTGGACCACGATGACCGGCTTTTTGCACG
CCTTTGCGCTGGCGCGCGCCGAGGGCGTCCCGGCAGAAGAGTTCGCACCGTTTTCGACTG
AAATGGCGTCGAGCATGCCAATTCTGGAAGAGATGGCCGAGCAAATCGACAGCGGTGAC
TACCCGGGTGACGAGAGCAACCTGGCTATGAATGCTGCTGCGATCGAACACATTGTTTCAC
GCAAGCCGTGATCAGGGTATCGATACCGCACTTCCGGAAGCGGTGAAAGCCCTCGCGGA
TCGTGCGATCGCCGCGGGCCATGGTGCGGACGGTTTCACCTCTCTGATTGAGGTCCTGCG
TAAACCGGCATAACTCGAG
```

Amino acid sequence of AltN559

```
MGSSHHHHHHSSGLVPRGSHMTNNTPTVTIGLGPMGQALAGFLAAGHPTTVWNRTPGKA
DELVARGAVRAATVAEAVAASPLVVVCVTDYDAMRAVLEPAADALAGRVLVNLTSDTPER
AREAAAWAAEHGVLDYLDGAIMVPPPVIGTPEAVVFYSGPREVFEAHRATLAALGGATYLGE
DPGLAALYDVALLDIFWTTMTGFLHAFALARAEGVPAEEFAPFATEMASSMPILEEMAEQID
SGDYPGDESNLAMNAAAIEHIVHASRDQGIDTALPEAVKALADRAIAAGHGADGFTSLIEVL
RKPA
```

AltN560

The ancestral gene encoding AltN560 was codon-optimized for overexpression in *E. coli* BL21 (DE3) using GenSmart Codon Optimization server (www.genscript.com/gensmart-free-gene-codon-optimization.html) and synthesized by GenScript Corporation (Nanjing, China). The gene was cloned into pET28(a) with T7 promoter, kanamycin resistance, and an N-terminal six-His-tag. These lower cases indicate amino acids of vector.

Codon-optimized gene sequence of AltN560

```
atgggcagcagccatcatcatcatcatcacagcagcgccctgggtgccgcggcgagccatATGACAAATAACACGCCCCGTT
ACTGTAATAGGTTTATAGGCCCTATGGGTCAGGCACTCGCAGGCGCGTTCCTGGCGGCGGGC
CATCCGACCACCGTTTGAACCGTACCCCGGGGAAGGCAGACGAGCTGGTGGCACGCGG
TGCGGTGCGCGCGGCGACGGTGGCCGAGGCGGTGGCGGCCTCCCCGCTGGTTGTGGTGT
GCGTTACCGATTACGATGCAATGCGTGCTGTGTTGGAACCGGCTGCTGACGCTCTGGCGG
GCCGTGTTTTGGTCAACTTGACCTCCGATACCCCGGAGCGCGCGAGAGAAGCCGCGGCCT
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GGGCAGCTGAGCACGGCGTAGATTACCTGGACGGCGCGATCATGGTTCCGCCACCGGTT
ATTGGCACCCCGGAAGCCGTGGTTTTCTATAGCGGTCCGCGTGAAGTTTTCGAGGCACAT
CGTGCTACGCTGGCGGCGCTGGGTGGTGCACATATTTGGGCGAAGATCCGGGTCTGGCA
GCGCTGTATGACGTTGCGTTGCTGGACATCTTTTGGACCACGATGACCGGCTTTTTGCACG
CCTTTGCGCTGGCGCGCGCCGAGGGCGTCCCGGCAGAAGAGTTCGCACCGTTTGC GACTG
AAATGGCGTCGAGCATGCCAATTCTGGAAGAGATGGCCGAGCAAATCGACAGCGGTGAC
TACCCGGGTGACGAGAGCAACCTGGCTATGAATGCTGCTGCGATCGAACACATTGTTTAC
GCAAGCCGTGATCAGGGTATCGATACCGCACTTCCGGAAGCGGTGAAAGCCCTCGCGGA
TCGTGCGATCGCCGCGGGCCATGGTGC GGACGGTTTACCTCTCTGATTGAGGTCCTGCG
TAAACCGGCATAACTCGAG

Amino acid sequence of AltN560

MGSSHHHHHSSGLVPRGSHMSNNQPVTVIGLGPMGQAMARAFLAAGHPVTVWNRTASKA
DELVAKGAVRAATVAEALAASELVVLSLTDYDAMYDVLEPAADALAGRVLVNLSSDTPER
AREAAAWAAEHGARYLTGGVMVPPPGIGTPEASIFYSGPREVFEAHRATLAVLGGTDYLGED
PGLAQLYYQAQLDLFWTTMTGYLHALALARAEGVPAEEFLPYATETADSMAFLEEMAAQID
SGEYPGDESNLAMMAASAEHIVHASRAAGIDTALPEAVKALFRRAVAAGHGADGFTSLIEVL
RKPA

Site-directed mutagenesis

The point variants of N559 and N560 were generated by site directed mutagenesis using pairs of complementary primers. The PrimeSTAR HS premix (Takara Biotechnology Company Limited, Dalian, China) was used to conduct these PCR experiments. These mutants were confirmed by sequencing (Tsingke Biotechnology Company, Shanghai, China).

Protein expression and purification

For heterologous overexpression of genes encoding ancestral IREDs, the expression vectors pET28(a) were transformed into competent *E. coli* BL21 (DE3) and grown overnight on selective Luria-Bertani (LB) agar plates containing 50 µg/mL kanamycin at 37 °C. The colonies first were inoculated LB medium containing 50 µg/mL kanamycin and grown overnight. One milliliter LB medium containing *E. coli* then was inoculated expression cultures in 100 mL terrific broth (TB) medium containing 50 µg/mL kanamycin. TB medium was placed at 37 °C in a shaking incubator for about 3 h, cooled down to 16 °C and keep for 10 min, induced with 0.5 mM isopropyl-β-D-thiogalactoside (IPTG), and subsequently grown overnight at 16 °C. Cells were harvested by centrifugation at 7826 x g for 5 min at 4 °C. For purification and solubility analysis, cell pellets were re-suspended in cold buffer A (20 mM NaPi, 500 mM NaCl, 10 mM imidazole, 5 mM beta-mercaptoethanol, pH 7.5). The resulting suspension was lysed using JY92-IIN ultrasonic homogenizer (Scientz, Ningbo, China) and the crude lysates were clarified by centrifugation at 7826 x g and 4 °C for 35 min. Clarified lysate supernatants were filtered (0.45 µm) and loaded onto a 10 mL Ni-NTA Beads 6FF (Smart-lifesciences, Changzhou, China) in a gravity column. The his₆-tagged ancestral protein was bound in Ni²⁺- affinity column and washed sequentially with 10 mL of buffer 1 (20 mM NaPi, 500 mM NaCl, 10 mM imidazole, 5 mM beta-mercaptoethanol, pH 7.5) and buffer 2 (20 mM NaPi, 500 mM NaCl, 108 mM imidazole, 5 mM beta-mercaptoethanol, pH 7.5) before elution of the protein using buffer 3 (20 mM NaPi, 500 mM NaCl, 250 mM imidazole, 5 mM beta-mercaptoethanol, pH 7.5). The fractions containing pure protein were concentrated using

Amicon Ultra-15 Centrifugal Filter (Merck Millipore) with a 10 kDa molecular weight cut-off membrane. The protein concentrate was exchange into buffer C (20 mM NaPi, 0.97 mM dithiothreitol, pH 7.5). Protein concentration was measured using NanoDrop 2000 Spectrophotometer (Thermo Fisher Scientific, Waltham, USA) using protein A208 function. All ancestral IREDs in this research were could be expressed as soluble. For protein crystallization, the concentrated fraction was subjected to size exclusion chromatography (SEC) using a HiLoad™ 16/600 Superdex™ 75 pg (Cytiva, Marlborough, USA) gel filtration column with buffer C as the eluent.

Activity assay

A UV-Vis spectrophotometer (Shimadzu, Kyoto, Japan) was used to determine ancestral enzyme activity toward imine substrates by monitoring the change in the absorbance of NADPH at 340 nm. Each 1mL reaction mixture contained NaPi buffer (100 mM, pH 7.0), 10 μ L NADPH (10 mM), 10 μ L substrate **1a** (final concentration for N1: 0.2 mM, 0.3 mM, 0.6 mM, 1 mM, 3 mM, 5 mM; N2: 0.01 mM, 0.03 mM, 0.1 mM, 0.3 mM, 1 mM, 2 mM, 5 mM; N3: 0.01 mM, 0.03 mM, 0.1 mM, 0.5 mM, 1 mM, 5 mM; N4: 0.01 mM, 0.02 mM, 0.05 mM, 0.1 mM, 1 mM; N5: 0.01 mM, 0.05 mM, 0.5 mM, 1 mM, 5 mM, 8 mM; N557: 0.1 mM, 0.5 mM, 1 mM, 2 mM, 3 mM; N558: 0.1 mM, 0.5 mM, 1 mM, 3 mM, 5 mM; N559: 0.1 mM, 0.5 mM, 1 mM, 3 mM, 5 mM; N560: 0.01 mM, 0.05 mM, 0.1 mM, 0.3 mM, 0.5 mM), and 0.1 mg/mL purified enzyme (final concentration). After measuring the initial absorbance of reaction mixtures, we placed them in a shaker (30 °C, 800 rpm) for 1 min, 2 min, 5 min or 10 min. At each time point, we measured the absorbance of the reaction mixture. The reaction time for N1, N2, N4, N5, and N557 was 2 min. The reaction time for N3 was 1 min. The reaction time for N558 and N560 was 5 min. N559 reacted for 10 min. The amount of enzyme that converts 1 μ mol substrate into product per minute was defined as one unit of activity.

Measurement of specific activity was also performed via monitoring the concentration change of product in the analytical-scale enzymatic reaction using GC. Each 500 μ L reaction mixture contained: 5 mM substrate **1a**, 1.25 mM NADPH, 0.5 mg/mL purified enzyme and 100 mM NaPi (pH 7.0). For N560, 1 mg/mL of purified enzymes were used in the reaction. Reactions were quenched by adding 20 μ L of 10 mM NaOH at 30 min for N1, N2, N3, AltN1, AltN2, AltN3, at 20 min for N4, N5, N557, N558, N559, N560, AltN4, AltN5, AltN557, AltN558, at 6 hours for AltN559 and at 24 hours for AltN560, respectively. The other reaction conditions are the same as the analytical-scale enzymatic reactions.

II. Protein crystallization experiment

Crystallization conditions screen. Sitting-drop vapor diffusion method was used to culture protein crystals at 18 °C. Appropriate crystallization conditions were screened using a range of commercially available conditions in 96-well plates. The crystallization mixture contains 1 μ L protein solution containing 2.5 mM NADP⁺ and 1 μ L precipitant reservoir solution. The protein crystallization of N559 and N559-M6 grown under the following two conditions. The condition one: 0.2 M NaCl, 0.1 M Bis-Tris (pH 5.5), 25% (v/v) PEG3350. The condition two: 0.5 M LiCl, 0.1 M Tris (pH 8.5), 28% (v/v) PEG6000. For obtaining high-quality diffraction, optimizing crystallization condition experiments were carried out. The best crystals were obtained in following conditions. For N559, the condition contains

0.1 M Bis-Tris (pH 5.5), 25% PEG3350 and 0.2 M NaCl at 18°C. For N559-M6, the best crystal grown was in 4 °C. The condition contains 0.1 M Bis-Tris (pH 6.5), 25% PEG3350 and 0.2 M NaCl.

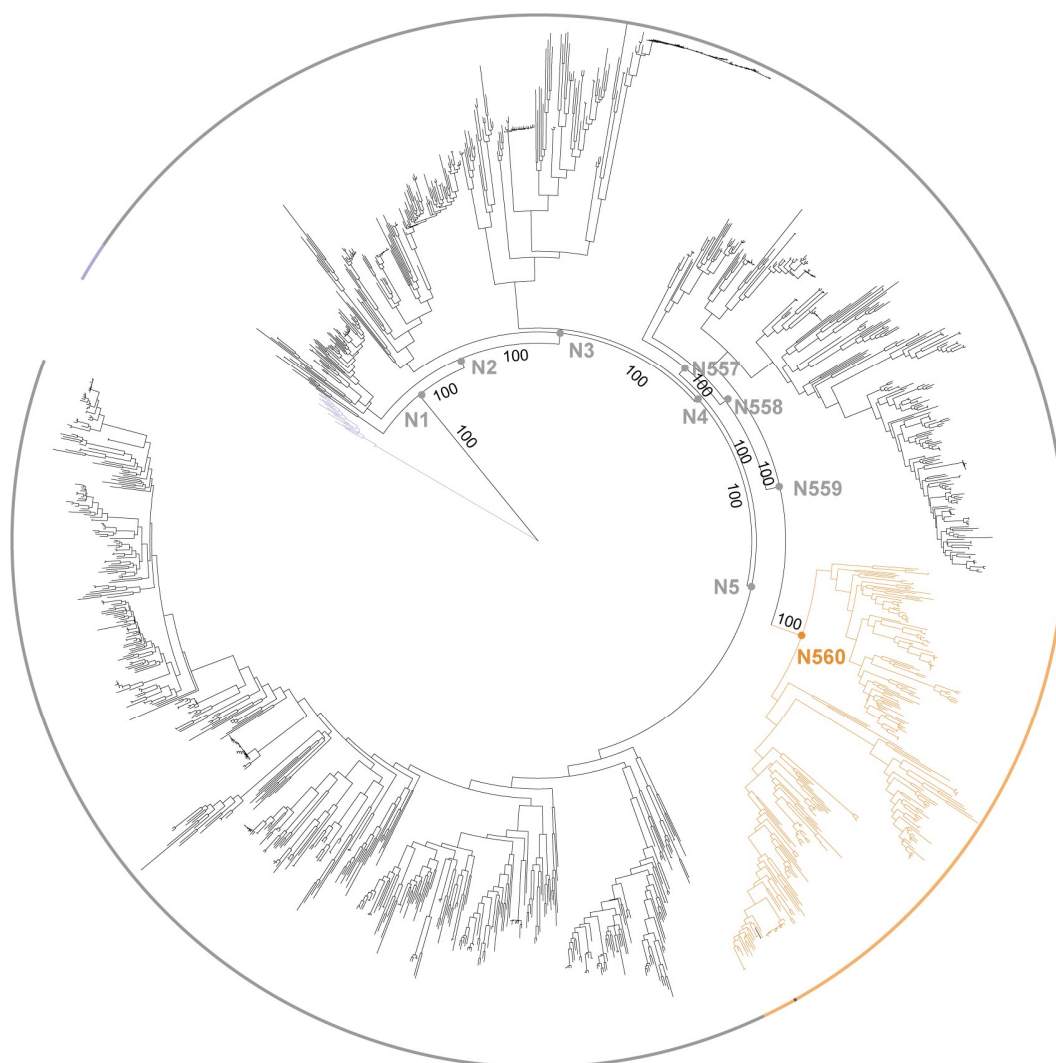
Data collection, structure solution, model building and refinement. All X-ray data were collected at beamline BL19U1 and BL02U1 of Shanghai Synchrotron Radiation Facility (SSRF). Diffraction data were indexed, integrated and scaled via the automatic processing pipeline at SSRF¹. The .mtz files outputted from the pipeline further were processed using Phenix². The space group of N559 and N559-M6 was P 21 21 21 and P 1 21 1, respectively. Molecular replacement method was used to solve protein structures. The initial model structure of N559 came from the prediction of Alphafold2, and the N559 structure was used as initial structure to solve N559-M6 structure. These structures were built and refined by a multiple iterative cycle using Coot³ and the Refinement tool in Phenix. Clear electron cloud density of NADP⁺ was observed in all structures. The two protein structures were finally validated upon submission to the PDB database. The PDB number of N559 and N559-M6 is 8JKU and 8HWY, respectively. These statistics of solution and refinement are presented in following table.

Molecular docking. Molecular docking was performed using autodock vina⁴. We define two distances used to evaluate whether the substrate is prereaction state. The distance between N atom of substrate and -OH of Asp/Tyr170, and the C atom of C=N in substrate and H of C4 in nicotinamide ring less than 5 Å.

III. GC and HPLC analysis: methods and conditions

Chiral gas chromatography column (CP-Chirasil Dex CB, 25 m × 0.25 mm × 0.25 μm) was used for determine the enantiomeric excess of products. The analytical method refers a pervious report⁶.

Normal phase liquid chromatography column (Chiralpak IA, 4.6 mm × 250 mm) was used for determined the enantiomeric excess of 1-phenyl-1,2,3,4-tetrahydroisoquinoline (**2f**). The following is method set: normal hexane (with 1% v/v diethylamine): isopropanol = 90:10, flow rate 0.8 mL/min, temperature 30 °C, detection wavelength 220 nm, and analysis time 22 min. The analytical method toward **2b**, **2c**, **2d**, and **2e** refers our previous publication⁵.



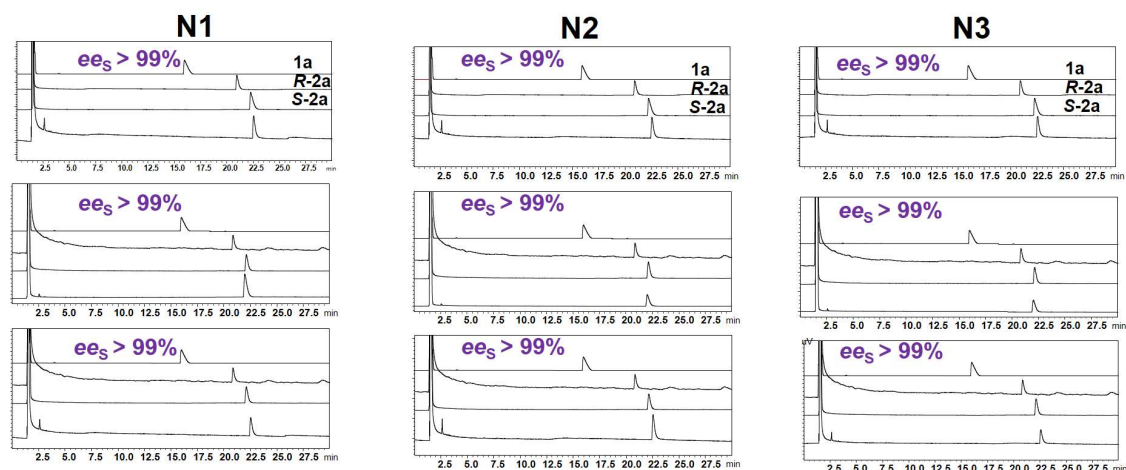
Supplementary Fig. 1

The phylogenetic tree of imine reductases. The yellow clade comprises all Y-type sequences, except for one sequence that is not Tyr at standard position 187, which is also included in this clade. The purple clade is the outgroup we select for determining the root of the ingroup. Moreover, the labels N1, N2, N3, N4, N5, N557, N558, N559, and N560 are reconstructed ancestral sequences in this research. The numbers in the branch are the transfer bootstrap expectation values calculated with BOOSTER (<https://booster.pasteur.fr/>). The phylogenetical tree was visualized using iTol (<https://itol.embl.de/>)⁸.

		130	140	150	160	170	180														
<i>SvlR</i>	125	MAIP	PDIA	TDAV	LLYSG	PKAA	FEHE	ATLR	ALGA	AGTT	YLD	TDH	GLS	ALY	MS	LLG	IM	WG	185		
<i>SelR</i>	125	LALP	QGIG	TDEA	TLLY	AGPK	AAFE	EHQA	TAVL	GEAA	TVYL	DE	DHGL	SALY	DM	AVL	TIM	WG	185		
<i>AalR3</i>	121	MAIP	SSIG	TAE	AVIL	LSGP	RP	DFD	- - - -	LGAL	- -	TVSH	LGED	PGLA	SLY	DV	AGL	TMM	WS	174	
<i>AdIR2</i>	120	MAIP	SSIG	TAE	AVIL	LSGP	RP	DFD	- - - -	LGAL	- -	TVSH	LGED	PGLA	SLY	DV	AGL	TMM	WS	173	
<i>StIR</i>	122	MITP	PGIG	TET	AVLF	YAGD	QAVF	DAHE	PVLK	LLG	-	AGTT	YLGT	DHGP	ALF	DV	SLL	GLM	WG	181	
<i>RIR</i>	122	MITP	PGIG	TET	AVLF	YAGT	QSVF	EKYE	PALK	LLG	-	GGTT	YLGT	DHGM	PALY	DV	SLL	GLM	WG	181	
<i>N1</i>	121	MATP	RLIG	TPEA	LILY	SGS	PEAFE	AHRE	TAA	LG	- -	DSTY	LGAD	PGLA	SLY	D	TALL	GLM	WG	179	
<i>N2</i>	121	MATP	PMIG	TPEA	LILY	SGS	PEAFE	AHRE	TAA	LG	- -	DSTY	LGAD	PGLA	SLY	D	TALL	GLM	WG	179	
<i>N3</i>	121	MATP	PMIG	TPEA	LILY	SGS	PEAFE	AHRE	TAA	LG	- -	GSTY	LGAD	PGLA	SLY	D	VALL	GLM	WG	179	
<i>N4</i>	121	MATP	PMIG	TPEA	LILY	SGS	REAFE	AHRE	TAA	LG	- -	GSTY	LGED	PGLA	ALY	D	VALL	GLM	WG	179	
<i>N5</i>	121	MATP	PMIG	TPEA	LILY	SGS	REVF	AHRE	TAA	LG	- -	GSTY	LGED	PGLA	ALY	D	VALL	GLM	WG	179	
<i>N557</i>	121	MATP	PMIG	TPEA	LILY	SGP	REVF	AHRE	TAA	LG	- -	GSTY	LGED	PGLA	ALY	D	VALL	GLM	WT	179	
<i>N558</i>	121	MVPP	PMIG	TPEA	LILY	SGP	REVF	AHR	TAA	LG	- -	GSTY	LGED	PGLA	ALY	D	VALL	GLM	WT	179	
<i>N559</i>	121	MVPP	PMIG	TPEA	LILY	SGP	REVF	AHR	TAA	LG	- -	GSTY	LGED	PGLA	ALY	D	VALL	DLF	FWT	179	
<i>N560</i>	121	MVPP	PGIG	TPEA	YVVF	YSGP	REVF	AHR	TAA	VLG	- -	GTDY	VGED	PGLA	QLY	YQA	QLD	I	FWT	179	
<i>SIR</i>	120	QVPP	PLIG	KPES	STYS	SGPK	DVFD	AHED	TLK	VLT	-	NADY	RGED	AGLA	AMY	YQA	QMT	I	FWT	178	
<i>ScIR</i>	121	TVPP	SGIG	SPES	SAFY	SGPS	AA	FERH	RE	TLT	L	-	RTDY	RGED	PGLA	ALM	YQ	IGM	V	MFWN	179

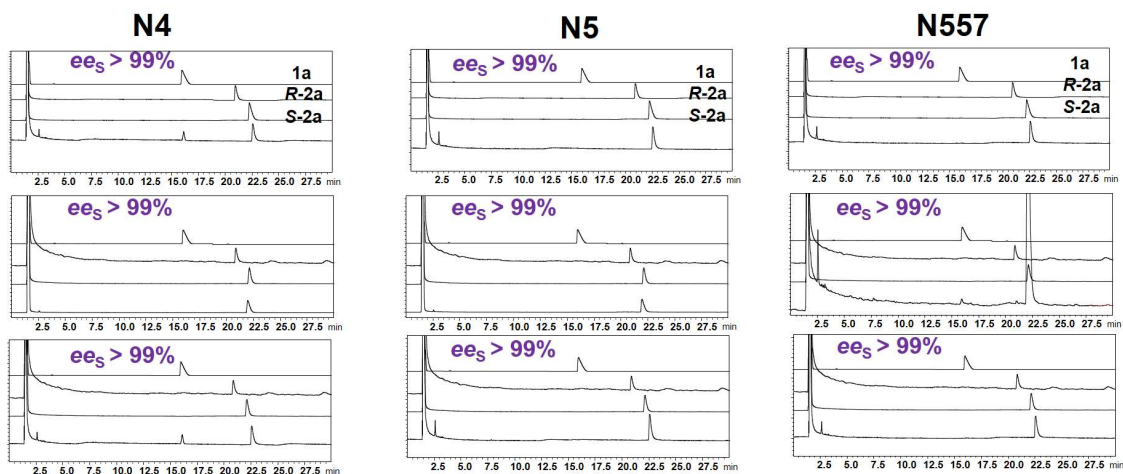
Supplementary Fig. 2

Multiple sequence alignment between ancestral and eight extant IRED sequences. The location highlighted with orange denotes the standard position 187 (according to the residue number in 3ZHB)⁷.



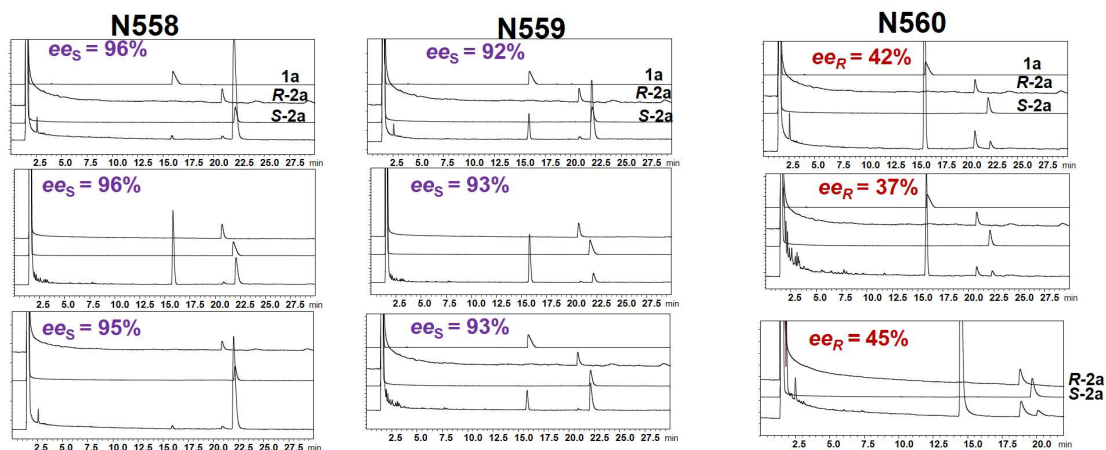
Supplementary Fig. 3

Chromatograms for the stereoselectivity analysis of N1–N3. The numbers on the graph denote the corresponding ee in each measurement. The experiments were independently conducted three times for each enzyme ($n = 3$).



Supplementary Fig. 4

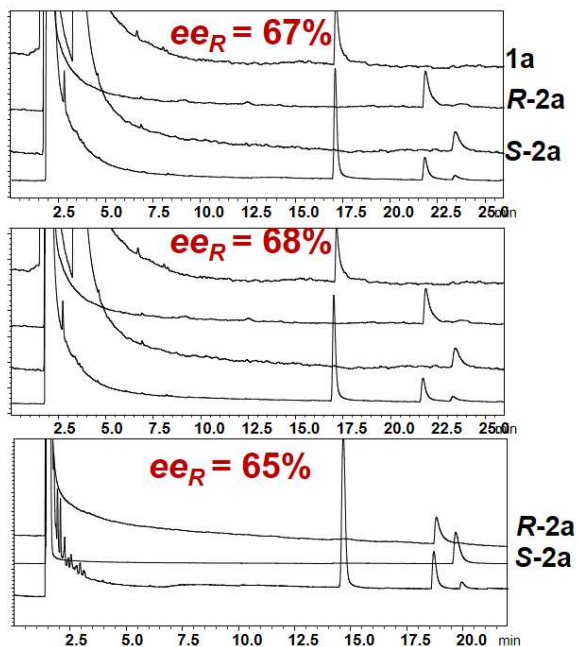
Chromatograms for the stereoselectivity analysis of N4–N557. The numbers on the graph denote the corresponding ee in each measurement. The experiments were independently conducted three times for each enzyme ($n = 3$).



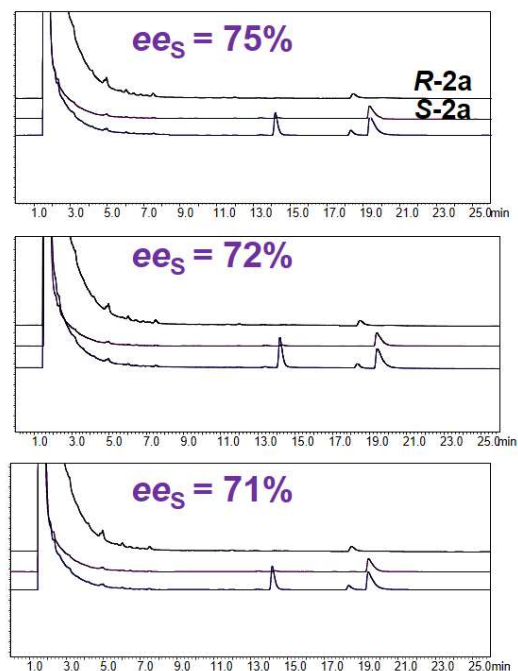
Supplementary Fig. 5

Chromatograms for the stereoselectivity analysis of N558–N560. The numbers on the graph denote the corresponding ee in each measurement. Some measurements showed different retention times due to the change in column efficiency during this study. The experiments were independently conducted three times for each enzyme ($n = 3$).

N559-M6



N560-M6



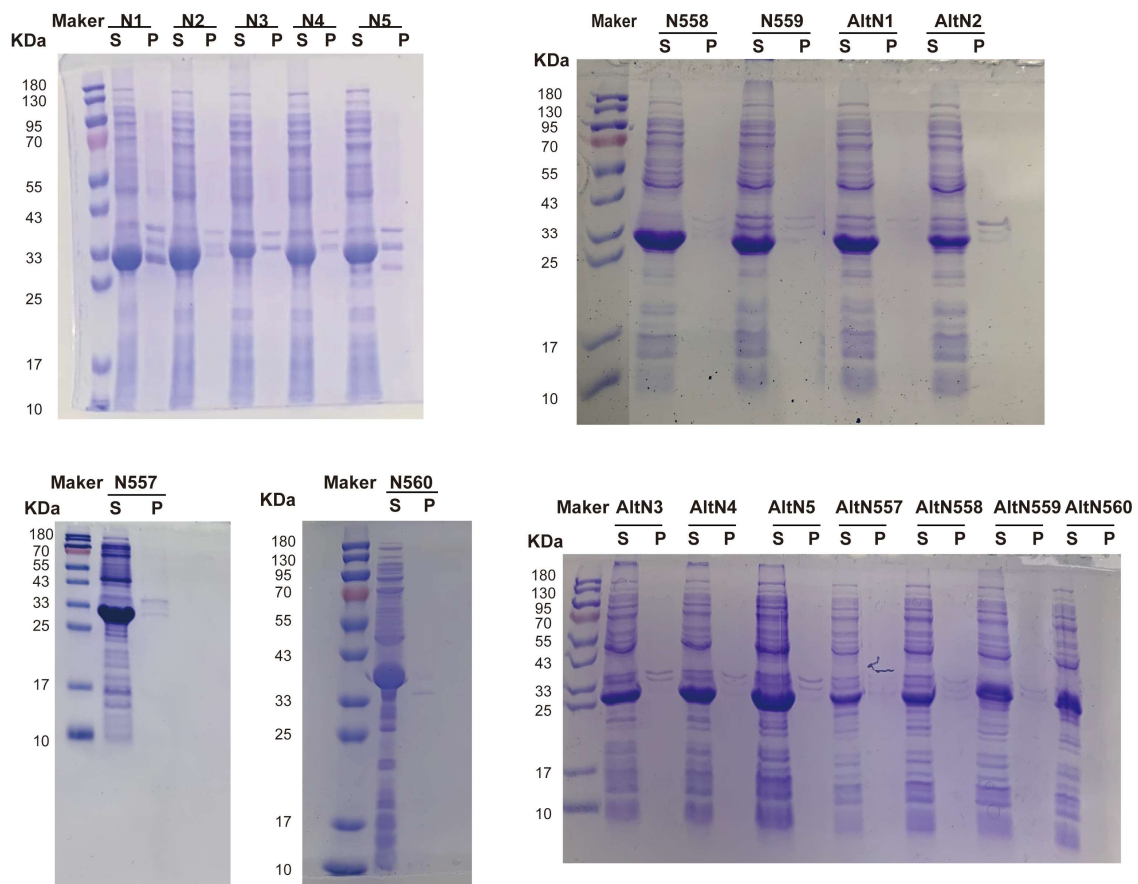
Supplementary Fig. 6

Chromatograms for the stereoselectivity analysis of N559-M6 and N560-M6. The numbers on the graph denote the corresponding ee in each measurement. The experiments were independently conducted three times for each enzyme ($n = 3$).

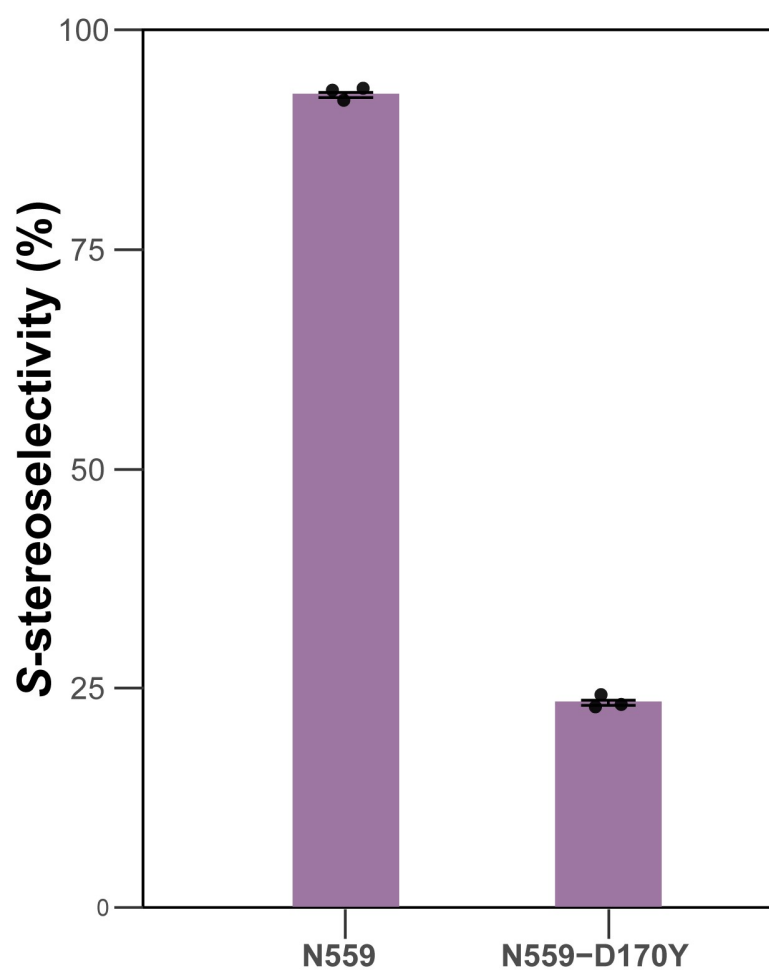


Supplementary Fig. 7

The stereoselectivity of alternative ancestral sequences toward substrate 1a. The red and yellow bars denote *S*- and *R*- enantiomers generated by ancestral IREDs, respectively. The *ee* of each alternative ancestral enzymes is obtained from one biotransformation reaction.

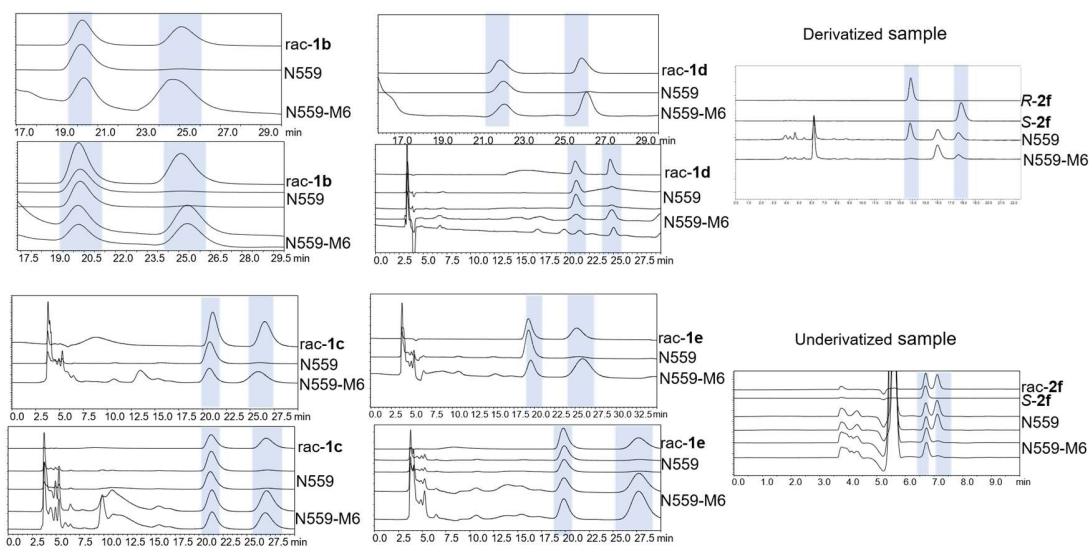


Supplementary Fig. 8
SDS-PAGE analysis of ancestral IREDs (N1–N560), and their alternative protein (AltN1–AltN560). The molecular weight of the proteins is between 29 to 35 kDa.



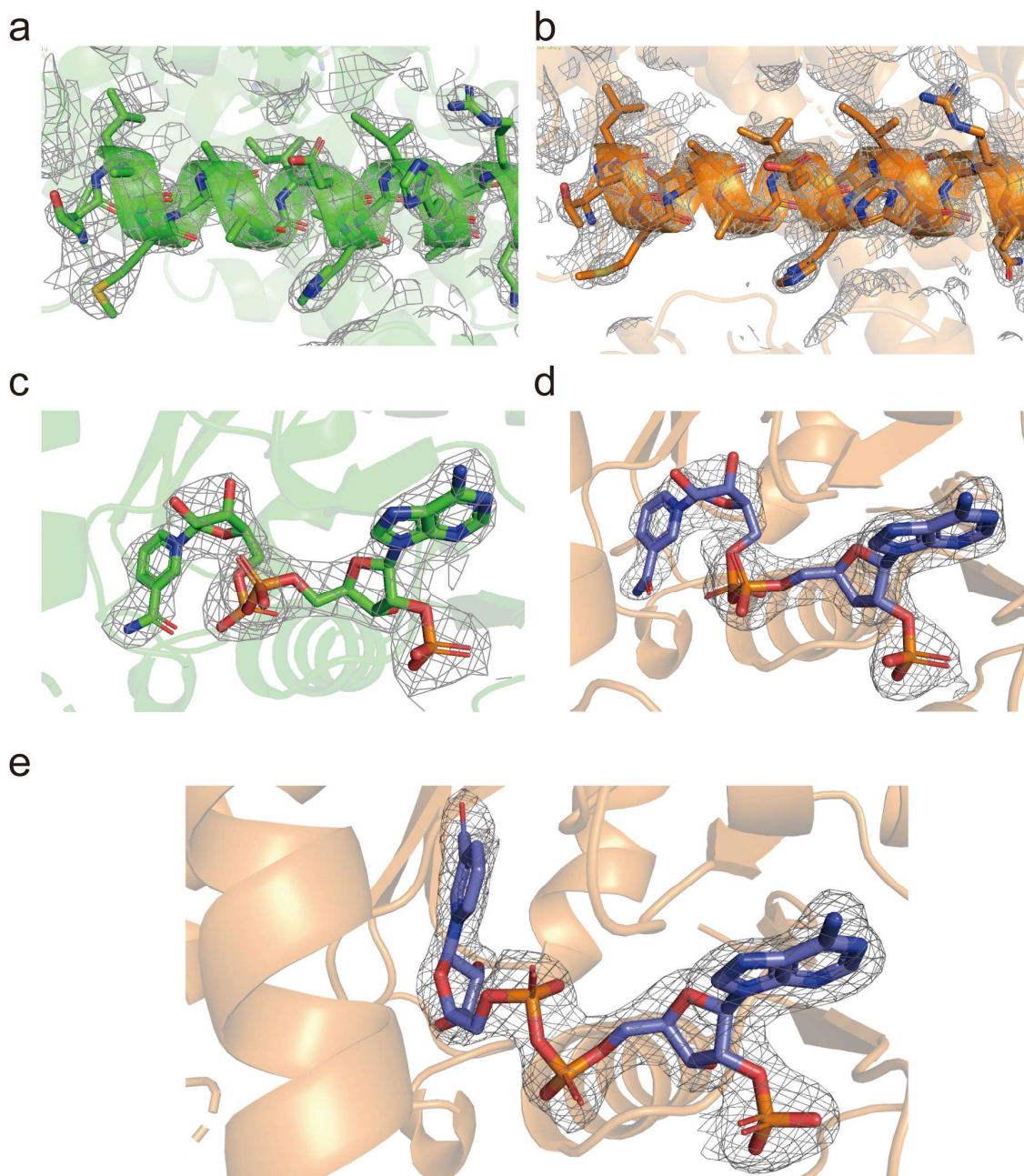
Supplementary Fig. 9

The ee of N559 and N559-D170Y toward substrate 1a. The ee values are shown as mean $\pm$ SEM calculated from three measurements ($n = 3$).



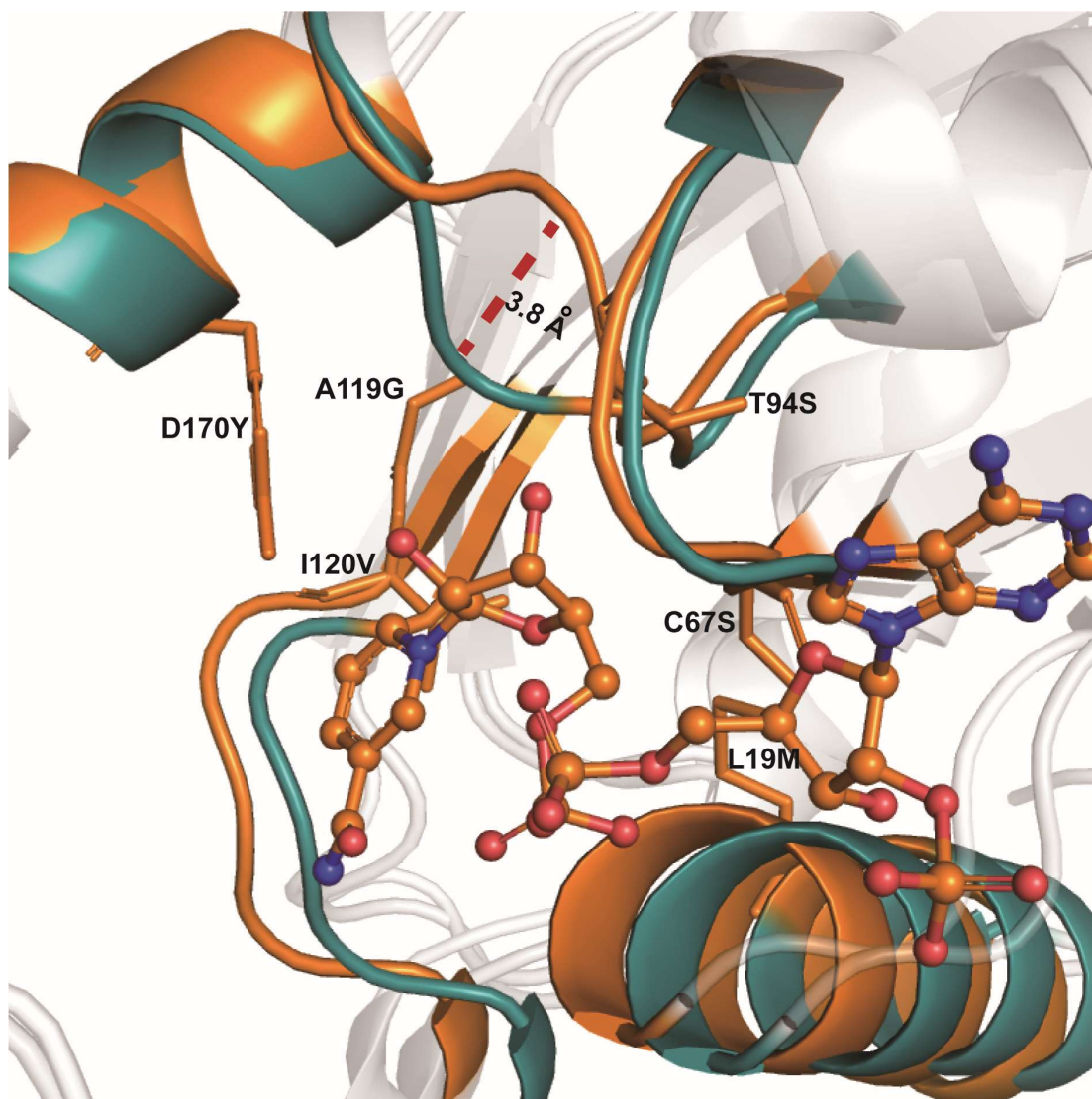
Supplementary Fig. 10

Chromatograms for the stereoselectivity analysis of N559 and N559-M6. *R*- and *S*-**2f** are products of **1f**. The product **2f** was derivatized by acetic anhydride, however, we found later that the configuration of **2f** could also be analyzed by liquid chromatography without derivatization. Therefore, **1f** has two types of chromatograms. The experiments were independently conducted three times for each enzyme ($n = 3$).



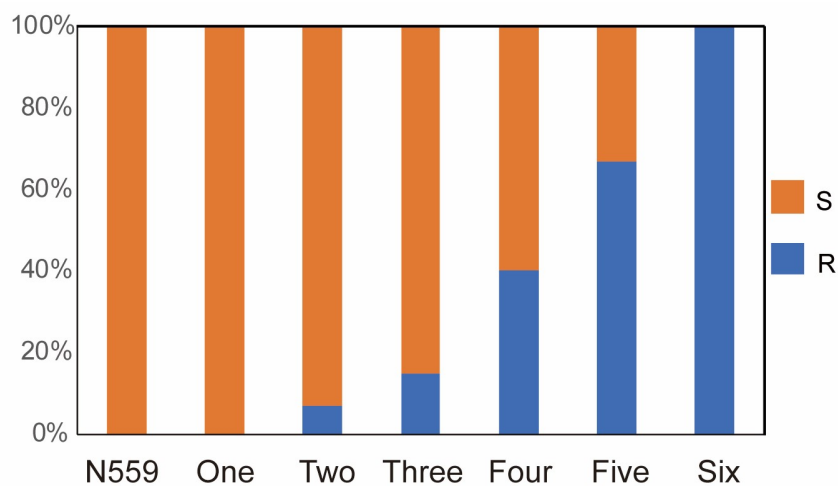
Supplementary Fig. 11

Omit maps of proteins and ligand-bound structures. **a** A portion of electron density map around NADP^+ 4 Å in N559. **b** A portion of electron density map around NADP^+ 4 Å in N559-M6. **c** Electron density map of NADP^+ in N559. **d** Canonical conformation of NADP^+ in the chain B of N559-M6. **e** A non-canonical conformation in IREDs in the chain A of N559-M6. NADP^+ is shown as ball-and-stick model in blue with the 2Fo-Fc electron density map contoured at the level of $\sigma = 1.0$.



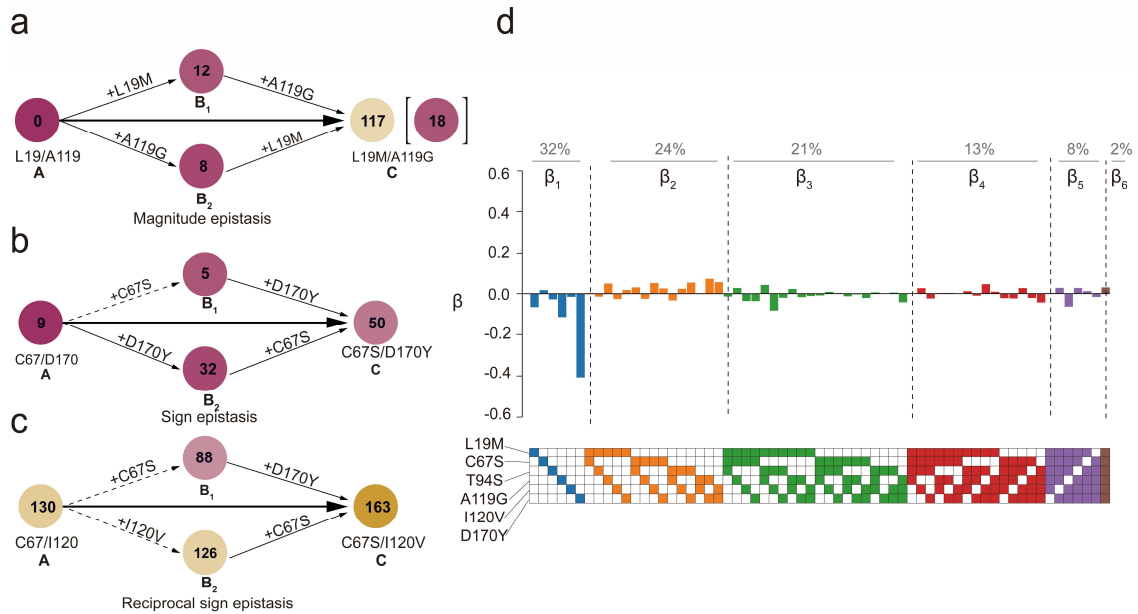
Supplementary Fig. 12

The conformational differences between N559 (deep teal) and N559-M6 (orange). The six historical mutations and NADP⁺ are shown as stick model in orange.



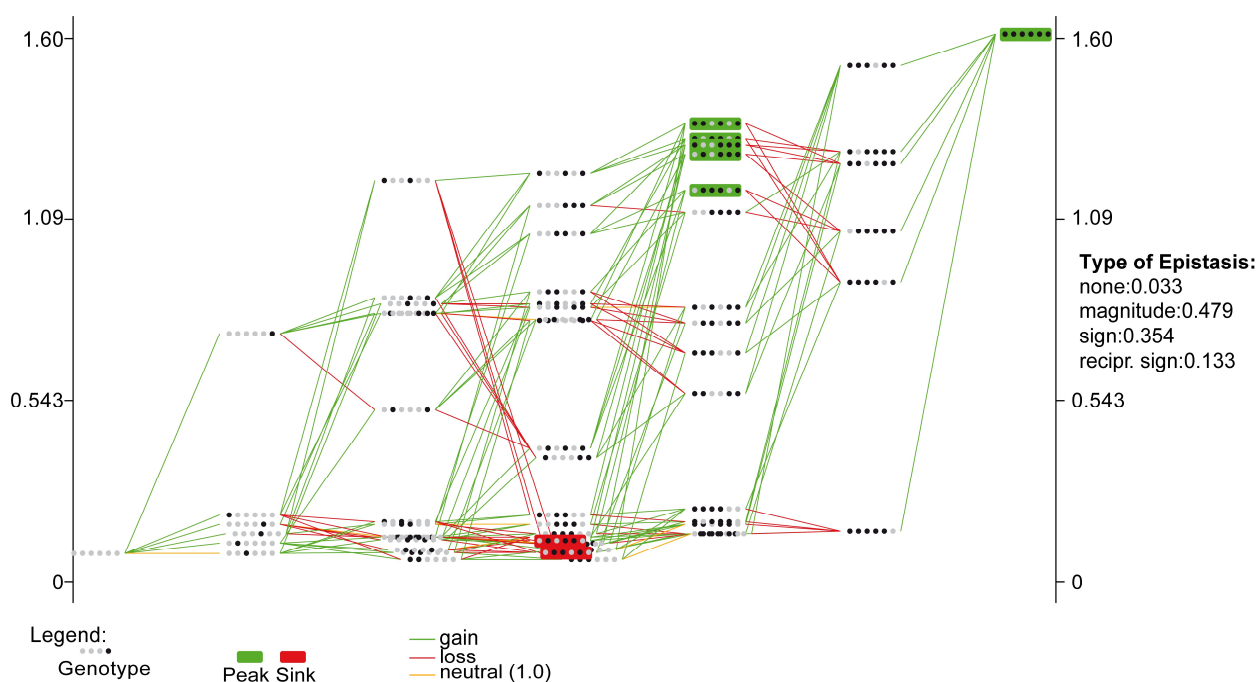
Supplementary Fig. 13

The proportion of mutants with *S*- and *R*- stereoselectivity. One, two, three, four, five, and six denote single, double, triple, quadruple, quintuple, and sextuple mutants, respectively. Yellow and blue bars denote *S*- and *R*-enantiomers generated by N559 and mutants, respectively.



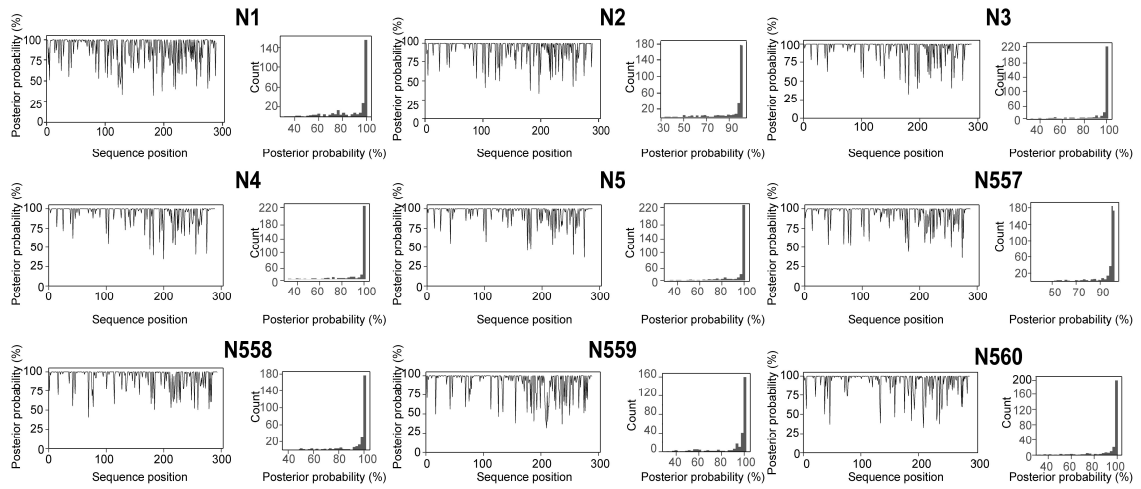
Supplementary Fig. 14

Investigation of epistasis in the empirical fitness landscape. a–c Selecting suitable mutations and intermediate sequences to illustrate epistasis types. **a** Magnitude epistasis: both two single mutations are beneficial to fitness, and the effect of a double mutant containing the two single mutations on fitness is greater than the simple additivity of the two single mutations. It only affects the magnitude of fitness. A: N559. The purple circle in the square bracket denotes a theoretical value resulting from the additive effect of L19M and A119G. The actual value of L19M/A119G is 117, which is greater than the theoretical value. Purple and yellow denote *S*- and *R*-stereoselectivity, respectively. The shade of color represents the magnitude of the numbers. **b** Sign epistasis: the effect of a single mutant is smaller than the fitness of the nonmutant and the double mutant, and the other single mutant is intermediate between the nonmutant and the double mutant. A: N559-L19M/I120V. **c** Reciprocal sign epistasis: the fitness of each single mutant is smaller than the fitness of the nonmutant and the double mutant. A: N559-L19M/T94S/A119G/D170Y. **d** Quantifying epistatic coefficients using the 'epistasis' Python package. Bars denote epistatic coefficient, magnitude and sign. Color denotes the order of epistasis: first (blue, β₁), second (yellow, β₂), third (green, β₃), fourth (red, β₄), fifth (purple, β₅), and sixth (brown, β₆). Filled square in the grid below the bars indicates the identity of mutations that contribute to coefficient. Y-axis is epistatic coefficient (β). The numbers above the bars denote the variations of function each order epistasis would explain.



Supplementary Fig.15

Statistical analysis of the empirical fitness landscape with the MAGELLAN tool. The Y-axis is the value quantifying the difference of fitness between N559 and each intermediate, and raw data are listed in Supplementary Table 5. The gain (Peak) denotes that the next node has more *R*-enantiomers in products compared to the previous node. The loss (Sink) denotes that the next node has less *R*-enantiomers in products compared to the previous node. The neutral line denotes that the next node has equal *R*-enantiomers in products compared to the previous node. These parameters used to analyze the landscape are default values in the MAGELLAN tool. Magnitude denotes magnitude epistasis. The sign and recipr. sign indicate sign and reciprocal epistasis, respectively.



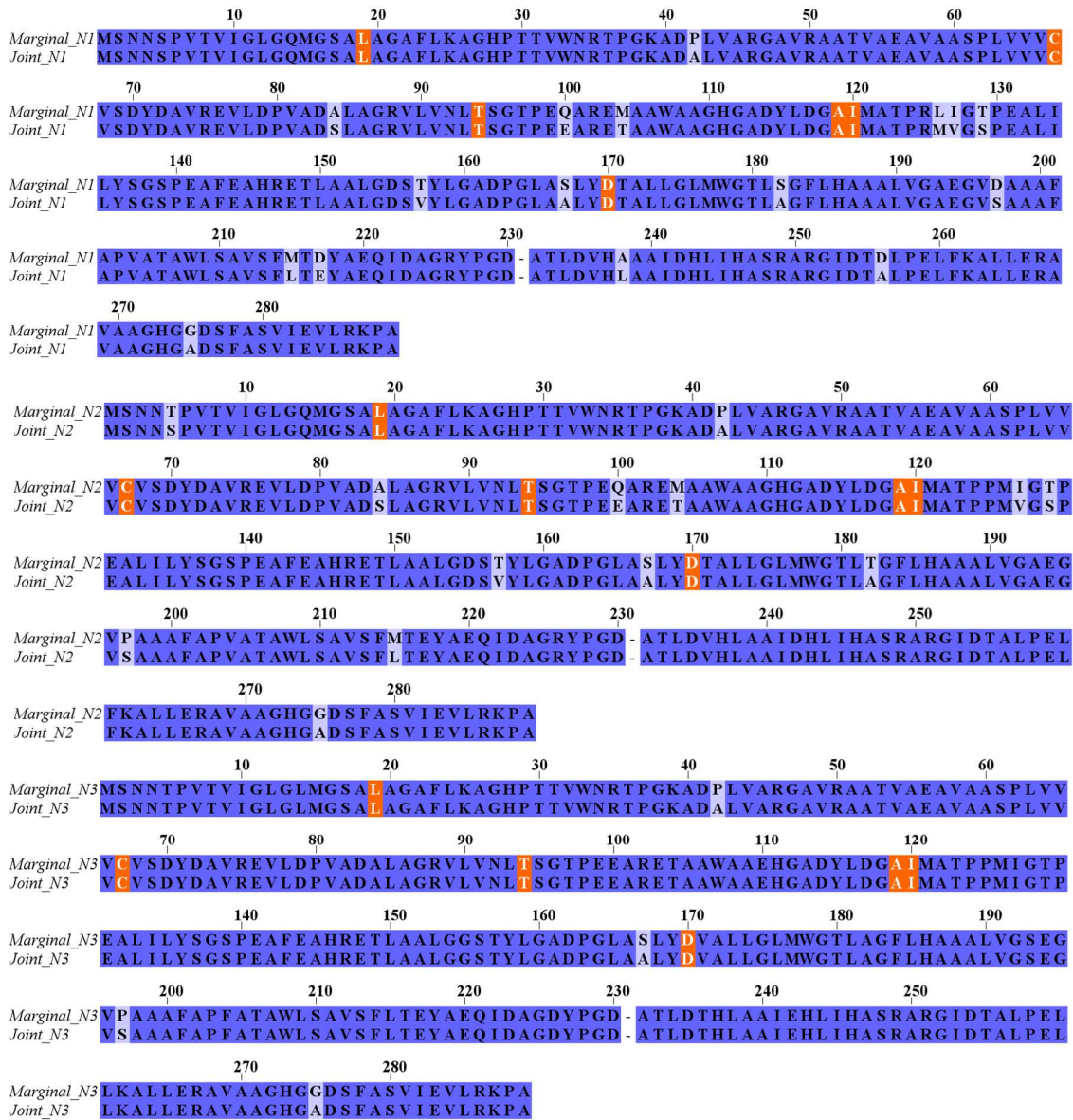
Supplementary Fig. 16

The distribution of maximum posterior probability for each site in all reconstructed ancestral sequences. Posterior probability with respect to position in the sequence is shown in a line chart. Distribution of posterior probabilities is shown on a histogram.

Enzyme	95	120	121	122	123	124	170	173	174	177	178	208	231	232	236
N1	S	I	M	A	T	P	D	L	L	M	W	W	-	A	V
N2	S	I	M	A	T	P	D	L	L	M	W	W	-	A	V
N3	S	I	M	A	T	P	D	L	L	M	W	W	-	A	T
N4	S	I	M	A	T	P	D	L	L	M	W	W	-	A	T
N5	S	I	M	A	T	P	D	L	L	M	W	W	-	S	M
N557	S	I	M	A	T	P	D	L	L	M	W	W	-	S	M
N558	S	I	M	V	P	P	D	L	L	M	W	W	E	S	M
N559	S	I	M	V	P	P	D	L	L	F	W	I	E	S	M
N560	S	V	M	V	P	P	Y	Q	L	F	W	M	E	D	M

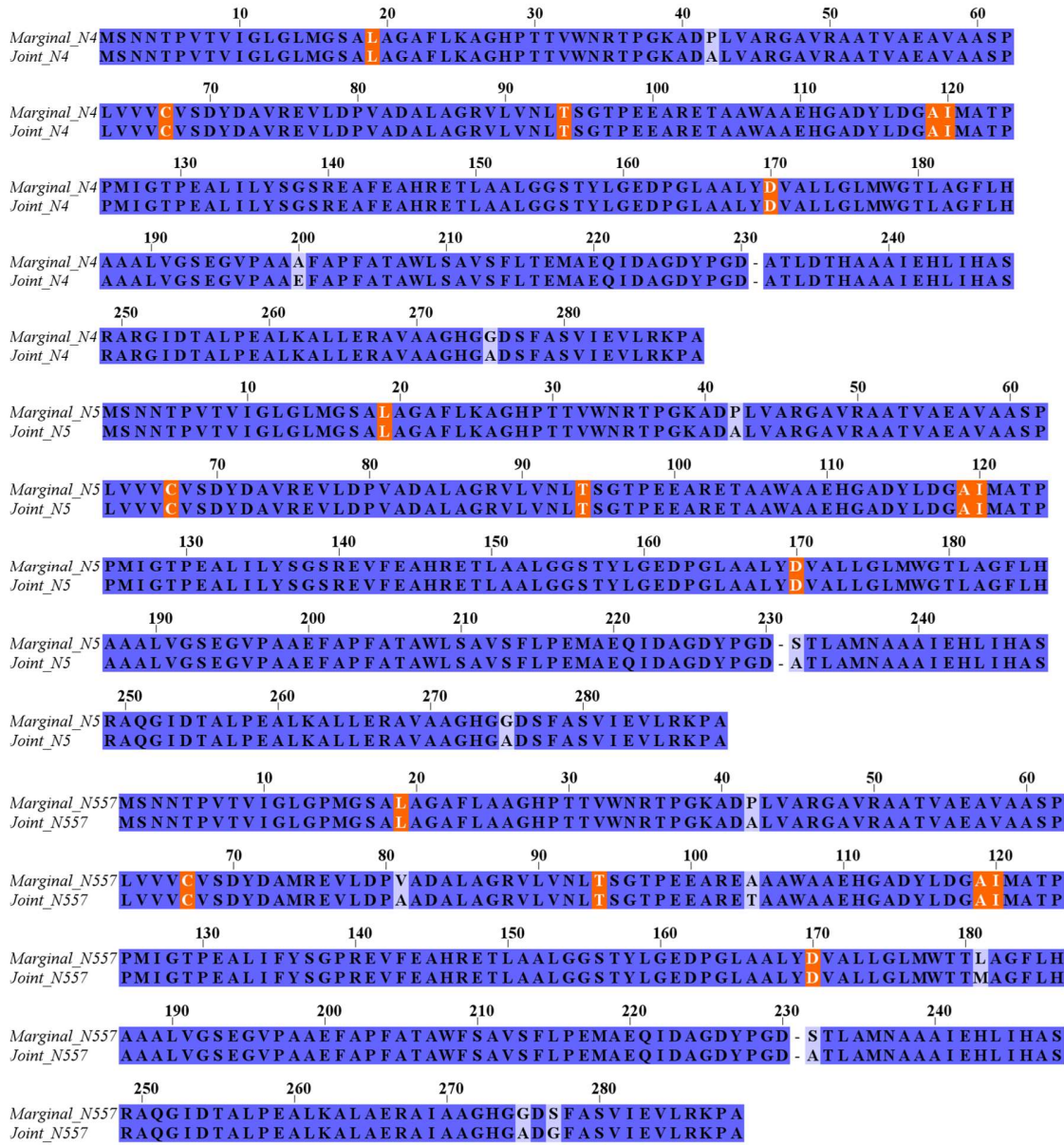
Supplementary Fig. 17

The residues around substrate 1a 6 Å in ancestral enzymes. The ‘-’ presents Indel mutations in multiple sequence alignment.



Supplementary Fig. 18

An examination of the sequence of marginal and joint reconstruction (N1–N3). Dark blue shade denotes identical sites, light blue shade represents the different sites between marginal and joint reconstruction, and orange shade denotes the six crucial historical substitutions in our manuscript.



Supplementary Fig. 19

An examination of the sequence of marginal and joint reconstruction (N4–N557). Dark blue shade denotes identical sites, light blue shade represents the different sites between marginal and joint reconstruction, and orange shade denotes the six crucial historical substitutions in our manuscript.



Supplementary Fig. 20

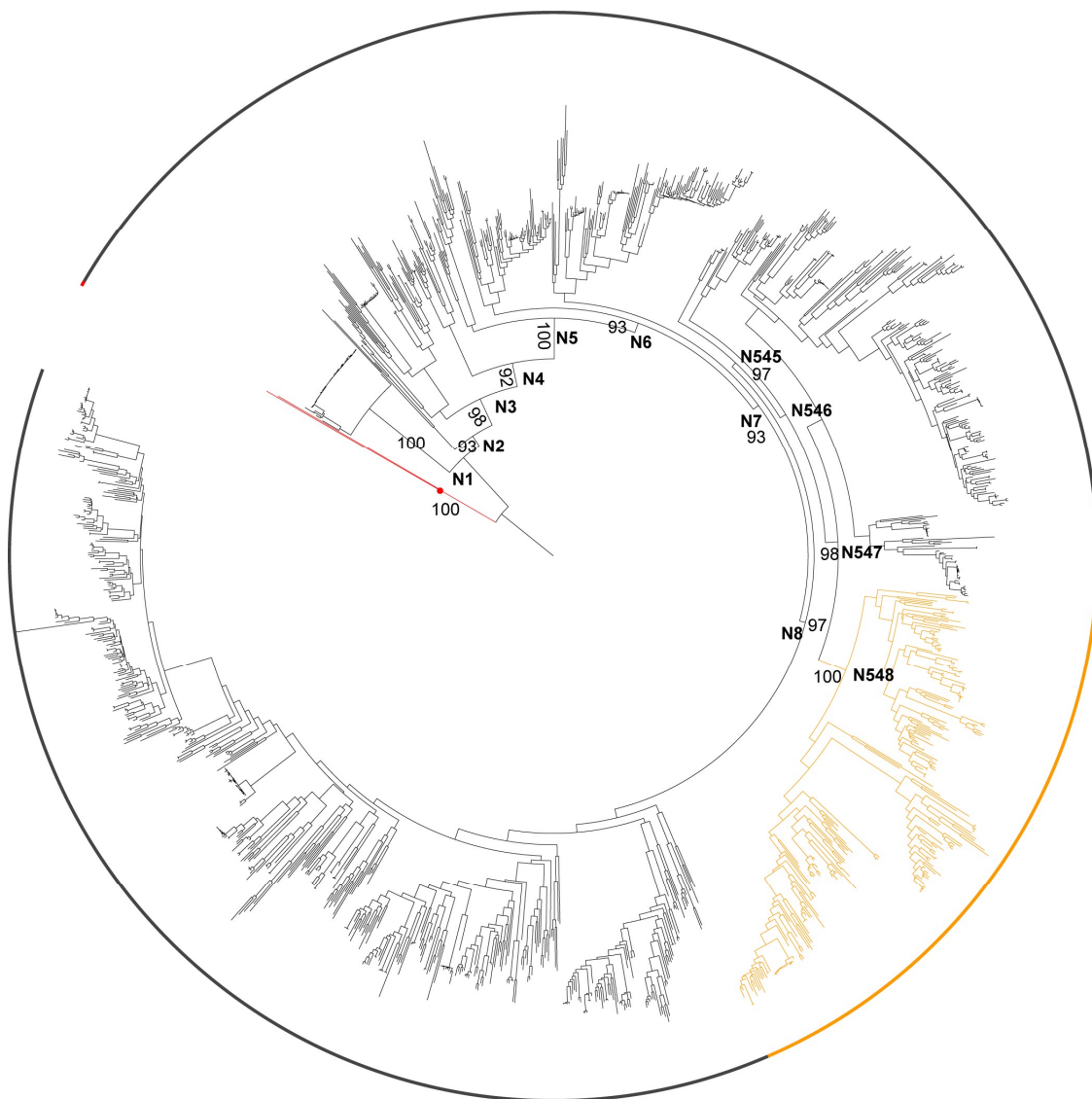
An examination of the sequence of marginal and joint reconstruction (N558–N560). Dark blue shade denotes identical sites, light blue shade represents the different sites between marginal and joint reconstruction, and orange shade denotes the six crucial historical substitutions in our manuscript.

Supplementary discussion

We have provided an alternative phylogenetic tree and corresponding ancestral sequences to further support the reliability of the conclusions in this manuscript.

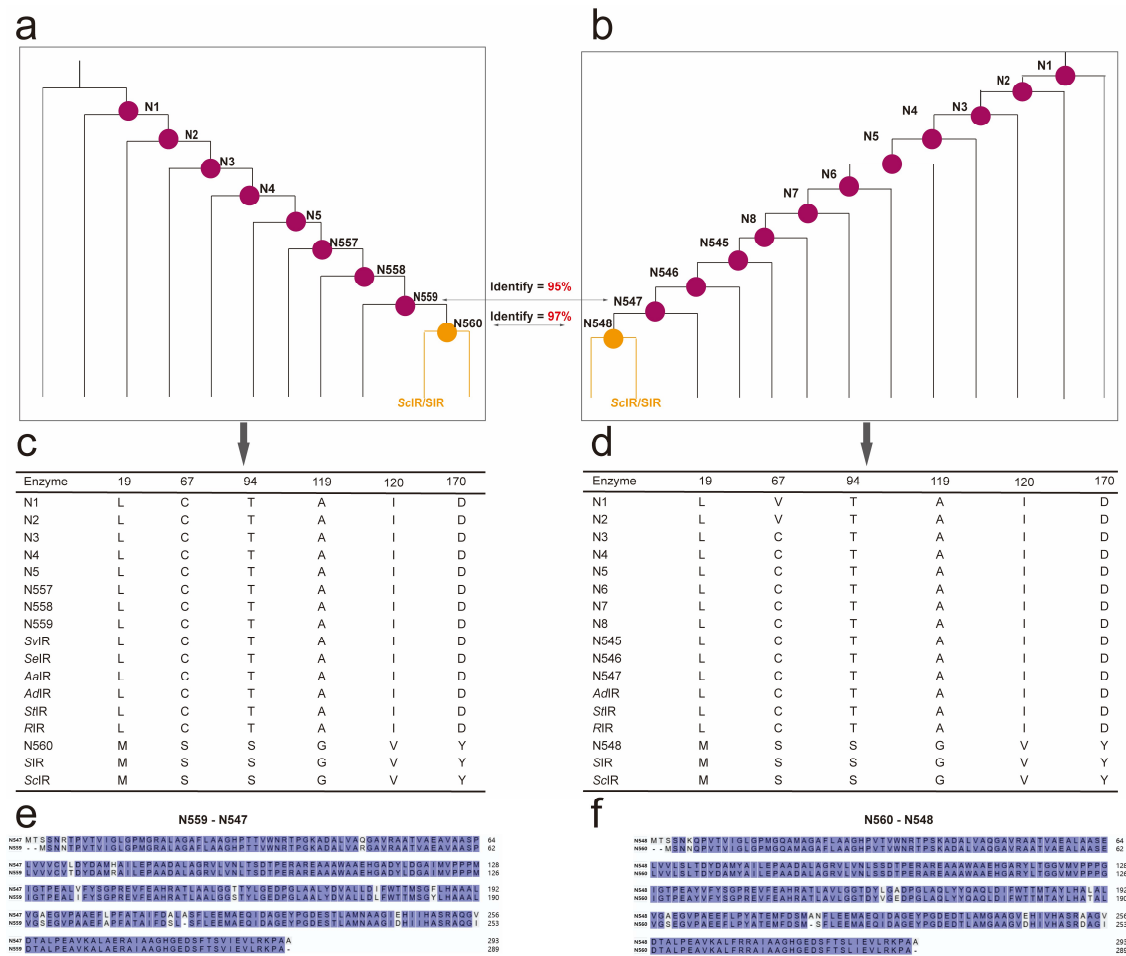
The biochemically characterized β -hydroxyacid dehydrogenases (β -HADs) and IREDs share considerable similarities in sequence and structure, indicating both enzyme families possess closest relative. Thus, we selected two β -HAD sequences as an outgroup to rebuilt the updated phylogenetic tree. Since the DASH function in the MAFFT server no longer supports the structure-based multiple sequence alignment (MSA) of more than 800 sequences, we performed the structure-based MSA using the PROMALS3D server (<http://prodata.swmed.edu/promals3d/promals3d.php>)⁹. The criteria for removing gaps remain unchanged from the previous description in the main manuscript. Then, we used IQtree (version 2.3.6 for Linux x86 64-bit) to the reconstruct phylogenetic tree with the following parameters: -m MFP -B 1000 -alrt 1000 -T AUTO. The best-fit model is chosen according to BIC: WAG + F + I + R10¹⁰⁻¹². The whole phylogenetic tree is present in Supplementary Fig. 21.

According to the updated phylogenetic tree, we found that the root is different from the tree in the main manuscript. However, the sequences containing Tyr at 170 still clustered in a clade (Yellow part in Supplementary Fig.21). We further reconstruct the ancestral sequences located in the focused evolution trajectory (from N1 to N548, corresponding to from N1 to N560 in the previous tree) with GRASP. Based on sequence analysis, the six historical mutations have not been affected by the root location (Supplementary Fig 22 c and d). Meanwhile, the sequences at the crucial evolution interval (N559-N560 and N547-N548) are highly conserved (Supplementary Fig 22 a, b, e, and f). More importantly, all protein engineering experiments were conducted for the N559–N560 evolution interval in this manuscript. Therefore, although the root location in the old tree may not be optimal, the new bioinformation analysis demonstrated that the findings in this manuscript are convincing.



Supplementary Fig. 21

The updated phylogenetical tree. The red branches denote the two outgroup sequences (PDB ID: 4GBJ and 3PEF). The yellow clade corresponds to the N560 clade of the old phylogenetic tree. NXs represent the node name, and the numbers are support values calculated using the ultrafast bootstrap approximation method in IQtree. These support values suggest that the phylogenetic tree is reliable.



Supplementary Fig. 22

Comparison analysis of ancestral sequences inferred from old and new phylogenetical trees. **a** The old simplified phylogenetical tree. **b** The new simplified phylogenetical tree. The red numbers denote sequence identify. **c** The six historical substitutions from previous ancestral sequence reconstruction. **d** The conservative analysis of the six sites from updated ancestral sequence reconstruction. **e–f** Sequence alignment of N559 vs N547 and N560 vs N548, respectively. Blue shadings indicate conservative sites, and grey shadings denote different residues at the sites.

Supplementary Table 1 Primer sequences.

Variants	Oligo sequences (5' to 3')
N559_D170Y	TTAGCCGCCCTGTATTATGTTGCACTGCTGGAT
N559_L19M	CCGATGGGTCGCGCAATGGCAGGCGCATTCTG
N559_C67S	CCGTTAGTTGTTGTTAGCGTGACCGATTATGAT
N559_T94S	GTGTTAGTTAATCTGAGCTCAGATACCCCGGAA
N559_A119G	GATTATCTGGATGGTGGCATTATGGTTCCGCCG
N559_I120V	TATCTGGATGGTGCAGTGATGGTTCCGCCGCCG
N560_Y170D	CTGGCCCAGCTGTATGATCAGGCACAGTTAGAT
N560_M19L	CCGATGGGTCAGGCACTGGCAGGCGCCTTTCTG
N560_S67C	GAATTAGTTGTGCTGTGCCTGACCGATTATGAT
N560_S94T	GTGTTAGTTAATCTGACCAGCGATACCCCGGAA
N560_G119A	CGCTATCTGACAGGAGCGGTTATGGTTCCGCCG
N560_V120I	TATCTGACAGGAGGCATTATGGTTCCGCCGCCG

Supplementary Table 2 The specific activity of each ancestor and corresponding alternative ancestral sequences from N1 to N560 toward **1a**.

Enzymes	Specific activity (mU/mg)
N1	56 ± 5
N2	46 ± 2
N3	43 ± 6
N4	46 ± 4
N5	60 ± 0.9
N557	62 ± 1
N558	30 ± 1
N559	1.4 ± 0.1
N560	14 ± 0.6
AltN1	38 ± 4
AltN2	20 ± 1
AltN3	53 ± 3
AltN4	39 ± 3
AltN5	80 ± 0.5
AltN557	18 ± 1
AltN558	60 ± 3
AltN559	4.2 ± 0.1
AltN560	0.3 ± 0.1
ScIR ^a	180 ± 0.3
SIR ^b	10

*Specific activity measured by gas chromatography. Data are shown as the mean ± SEM calculated from three measurements (n = 3). ^aThe specific activity of extant enzymes was quoted from our previous report⁵. ^bWe obtained the data from a master's thesis, which served as the foundation for our previous publication⁵. AltN1–AltN5 and AltN557–AltN560 are the alternative ancestor versions of N1–N5 and N557–N560, respectively.

Supplementary Table 3 Stereoselectivities of N559 and N559-M6 toward **1b–1f**.

Substrates	Enzymes	<i>ee</i> (%)		
1b	N559	89	90	89
	N559-M6	−40	−42	−41
1c	N559	86	87	88
	N559-M6	−19	−17	−18
1d	N559	84	89	86
	N559-M6	−29	−32	−24
1e	N559	84	90	82
	N559-M6	−41	−43	−48
1f	N559	−17	−18	−25
	N559-M6	88	89	80

*The negative numbers denote *R*-stereoselectivity. Due to structural similarity, the products' stereoselectivity of **1b–1e** referred to **1a** and our previous publication⁵.

Supplementary Table 4 Data collection and refinement statistics of N559 and N559-M6

	N559 complex with NADP ⁺	N559-M6 complex with NADP
Data collection		
Space group	P 21 21 21	P 1 21 1
Cell dimensions		
<i>a</i> , <i>b</i> , <i>c</i> (Å)	71.676 76.167 104.962	60.3359 71.7617 65.1075
α , β , γ (°)	90 90 90	90 113.202 90
Resolution (Å)	61.65–2.57 (2.662–2.57)	31.07–2.32 (2.403–2.32)
<i>R</i> _{merge}	0.1399 (1.009)	0.06837 (0.1655)
<i>I</i> / σI	11.65 (2.38)	15.50 (6.38)
Completeness (%)	95.83 (96.91)	99.24 (93.09)
Redundancy	6.1 (6.3)	5.9 (4.0)
Refinement		
Resolution (Å)	61.646–2.568 (2.660–2.568)	52.201–2.320 (2.403–2.320)
No. reflections	18150 (1793)	22065 (2048)
<i>R</i> _{work} / <i>R</i> _{free}	0.1969/0.2592	0.1696/0.2246
No. atoms		
Protein	4103	4127
Ligand/ion	146	146
Water	56	137
<i>B</i> -factors		
Protein	51.40	32.34
Ligand/ion	52.30	40.83
Water	44.03	34.43
R.m.s. deviations		
Bond lengths (Å)	0.014	0.014
Bond angles (°)	1.03	1.15

*One crystal was used for each structure. *Values in parentheses are for highest-resolution shell.

Supplementary Table 5 The enantiomer excess and specific activity of each intermediate in adaptive landscape

	<i>ee</i> (%)	Specific activity (mU/mg)
Start point		
N559	93 ± 0.3	1.40 ± 0.10
Single point mutation		
L19M	83 ± 0.3	1.14 ± 0.04
C67S	92 ± 0.3	0.56 ± 0.20
T94S	86 ± 0.5	1.52 ± 0.04
A119G	87 ± 0.2	1.70 ± 0.10
I120V	86 ± 0.6	0.90 ± 0.08
D170Y	26 ± 1	0.39 ± 0.03
Two-point mutations		
L19M/C67S	89 ± 0.4	1.52 ± 0.10
L19M/T94S	85 ± 0.3	3.09 ± 0.30
L19M/A119G	−22 ± 1	0.62 ± 0.04
L19M/I120V	86 ± 0.2	0.92 ± 0.06
L19M/D170Y	18 ± 1	0.53 ± 0.04
C67S/T94S	89 ± 0.14	1.09 ± 0.04
C67S/A119G	88 ± 0.3	1.36 ± 0.20
C67S/I120V	90 ± 0.2	0.90 ± 0.03
C67S/D170Y	50 ± 0.3	0.30 ± 0.02
T94S/A119G	86 ± 0.4	2.08 ± 0.10
T94S/I120V	89 ± 0.5	1.77 ± 0.02
T94S/D170Y	15 ± 0.3	0.69 ± 0.05
A119G/I120V	90 ± 0.2	1.99 ± 0.20
A119G/D170Y	15 ± 2	0.62 ± 0.08
I120V/D170Y	20 ± 0.2	0.90 ± 0.01
Three-point mutation		
L19M/C67S/T94S	87 ± 0.2	1.41 ± 0.05
L19M/C67S/A119G	86 ± 0.5	0.75 ± 0.08
L19M/C67S/I120V	90 ± 0.2	0.92 ± 0.004
L19M/C67S/D170Y	22 ± 0.7	0.43 ± 0.05
L19M/C94S/A119G	83 ± 0.4	1.78 ± 0.04
L19M/T94S/I120V	86 ± 0.5	1.97 ± 0.13
L19M/T94S/D170Y	17 ± 0.4	0.49 ± 0.02
L19M/A119G/I120V	88 ± 0.7	1.30 ± 0.01
L19M/A119G/D170Y	−24 ± 0.3	0.50 ± 0.01
L19M/I120V/D170Y	63 ± 0.9	0.26 ± 0.02
C67S/T94S/A119G	87 ± 1	1.09 ± 0.10
C67S/T94S/I120V	93 ± 0.6	2.38 ± 0.30
C67S/T94S/D170Y	13 ± 1	0.26 ± 0.01
C67S/A119G/I120V	91 ± 0.02	1.04 ± 0.08
C67S/A119G/D170Y	62 ± 2	0.47 ± 0.04
C67S/I120V/D170Y	20 ± 0.07	0.30 ± 0.02

T94S/A119G/I120V	86 ± 0.2	3.02 ± 0.10
T94S/A119G/D170Y	-5 ± 2	0.54 ± 0.02
T94S/I120V/D170Y	18 ± 0.2	0.61 ± 0.02
A119G/I120V/D170Y	-14 ± 0.6	0.49 ± 0.04
Four-point mutation		
L19M/C67S/T94S/A119G	81 ± 1	2.18 ± 0.20
L19M/C67S/T94S/I120V	87 ± 3	1.62 ± 0.20
L19M/C67S/T94S/D170Y	32 ± 5	0.35 ± 0.01
L19M/C67S/A119G/I120V	86 ± 1.2	1.81 ± 0.20
L19M/C67S/A119G/D170Y	-40 ± 0.5	0.66 ± 0.01
L19M/C67S/I120V/D170Y	45 ± 1	0.40 ± 0.02
L19M/T94S/A119G/I120V	85 ± 1	2.30 ± 0.05
L19M/T94S/A119G/D170Y	-35 ± 1	0.77 ± 0.09
L19M/T94S/I120V/D170Y	18 ± 0.6	0.72 ± 0.02
L19M/A119G/I120V/D170Y	-33 ± 2	1.06 ± 0.003
C67S/T94S/A119G/I120V	89 ± 0.4	1.58 ± 0.03
C67S/T94S/A119G/D170Y	-19 ± 0.6	0.36 ± 0.30
C67S/T94S/I120V/D170Y	23 ± 0.7	0.21 ± 0.01
C67S/A119G/I120V/D170Y	-30 ± 0.3	0.21 ± 0.0009
T94S/A119G/I120V/D170Y	-12 ± 0.01	0.58 ± 0.01
Five-point mutation		
L19M/C67S/T94S/A119G/I120V	88 ± 0.4	1.61 ± 0.02
L19M/C67S/T94S/A119G/D170Y	10 ± 0.1	0.61 ± 0.05
L19M/C67S/T94S/I120V/D170Y	-58 ± 0.2	0.23 ± 0.01
L19M/C67S/A119G/I120V/D170Y	-27 ± 0.8	0.20 ± 0.01
L19M/T94S/A119G/I120V/D170Y	-31 ± 0.01	0.51 ± 0.07
C67S/T94S/A119G/I120V/D170Y	-6 ± 0.6	0.57 ± 0.01
Six-point mutation		
L19M/C67S/T94S/A119G/I120V/D170Y	-67 ± 0.8	0.78 ± 0.07

*The positive number denotes *S*-stereoselectivity, and the negative number represents *R*-stereoselectivity. Data are shown as the mean ± SEM calculated from three measurements (n = 3).

Supplementary Table 6 Correlation analysis between the numbers of mutation and the proportion of mutants with *R*-stereoselectivity.

Number of mutations	Proportion of mutant with <i>R</i> -stereoselectivity ^a	Correlation coefficient ^b	t	p
0	0			
1	0			
2	0.070			
3	0.15	0.94	5.80	0.0020
4	0.40			
5	0.67			
6	1			

^aThe proportion of mutant with *R*-stereoselectivity = The number of mutant with *R*-stereoselectivity/Total number of corresponding mutant set. For example, 0 = 0/6 (six single mutants), 0.07 = 1/15 (fifteen double mutants), 0.15 = 3/20 (twenty triple mutants) and so on. ^bThe correlation analysis was performed by the Pearson correlation method with the Cor.test function in R package¹³, and the statistical test was two-sided.

Supplementary Table 7 Kinetic parameters of the ancestral enzymes toward model substrate **1a**.

Enzymes	K_m (mM)	k_{cat} (s^{-1})	k_{cat}/K_m ($s^{-1} mM^{-1}$)
N1	0.42 ± 0.06	0.039 ± 0.0003	0.094 ± 0.007
N2	0.030 ± 0.001	0.052 ± 0.003	1.6 ± 0.03
N3	0.041 ± 0.001	0.070 ± 0.001	1.7 ± 0.04
N4	0.013 ± 0.002	0.026 ± 0.0007	2.2 ± 0.34
N5	0.25 ± 0.025	0.034 ± 0.0007	0.14 ± 0.02
N557	2.96 ± 0.61	0.059 ± 0.007	0.02 ± 0.003
N558	1.73 ± 0.2	0.060 ± 0.007	0.016 ± 0.01
N559	2.13 ± 0.09	0.0074 ± 0.0002	0.0034 ± 0.0001
N560	0.11 ± 0.007	0.011 ± 0.0004	0.098 ± 0.004
ScIR ^a	0.17	0.16	0.94
SvIR ^a	0.064	8.40	131
SlIR ^a	0.044	0.53	12.10
RIR ^a	0.086	0.77	8.90

^aKinetic parameters cited from our previous report⁵. k_{cat} denotes catalytic constant. K_m represents Michaelis-Menton constant.

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